

Clinical proof-of-concept

ECG anomaly detection — example beats

Model: mitbih_autoencoder_best.pt
Threshold (F1-optimal, real-time preprocessing): 0.0434
Records used: 100,101,102,103,104,105,106,107
Total beats inspected: 17355 Anomalies in set: 6890

Confusion matrix on these beats:

True positives	: 6605	(correctly caught)
False negatives	: 285	(missed by the model)
False positives	: 250	(false alarms)
True negatives	: 10215	(correctly cleared)

Sensitivity = 0.959

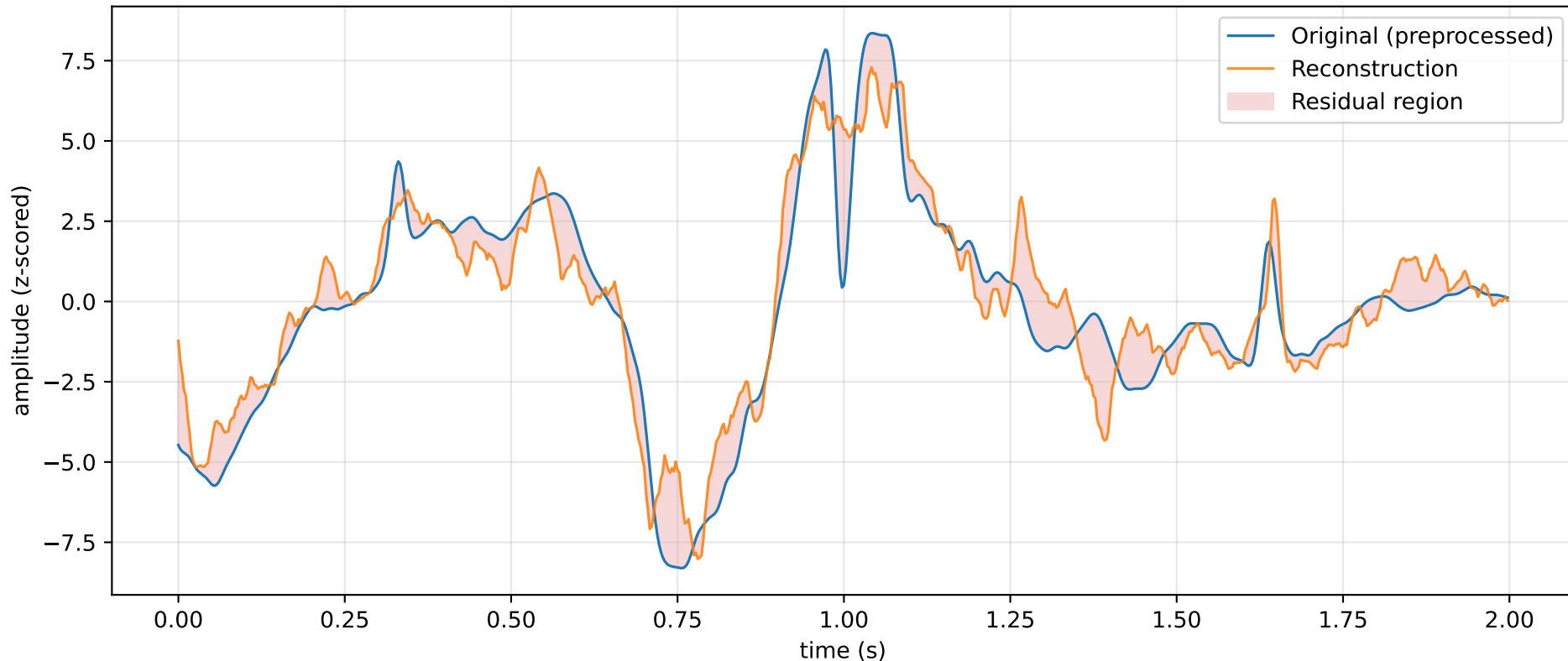
Specificity = 0.976

How to read the next pages: each page shows one 2-second window with the original ECG (blue) and the model's reconstruction (orange). The shaded red area between them is the reconstruction error. A beat is flagged when that error exceeds the threshold above.

TRUE POSITIVE — model correctly flagged abnormal

(12 example(s) follow)

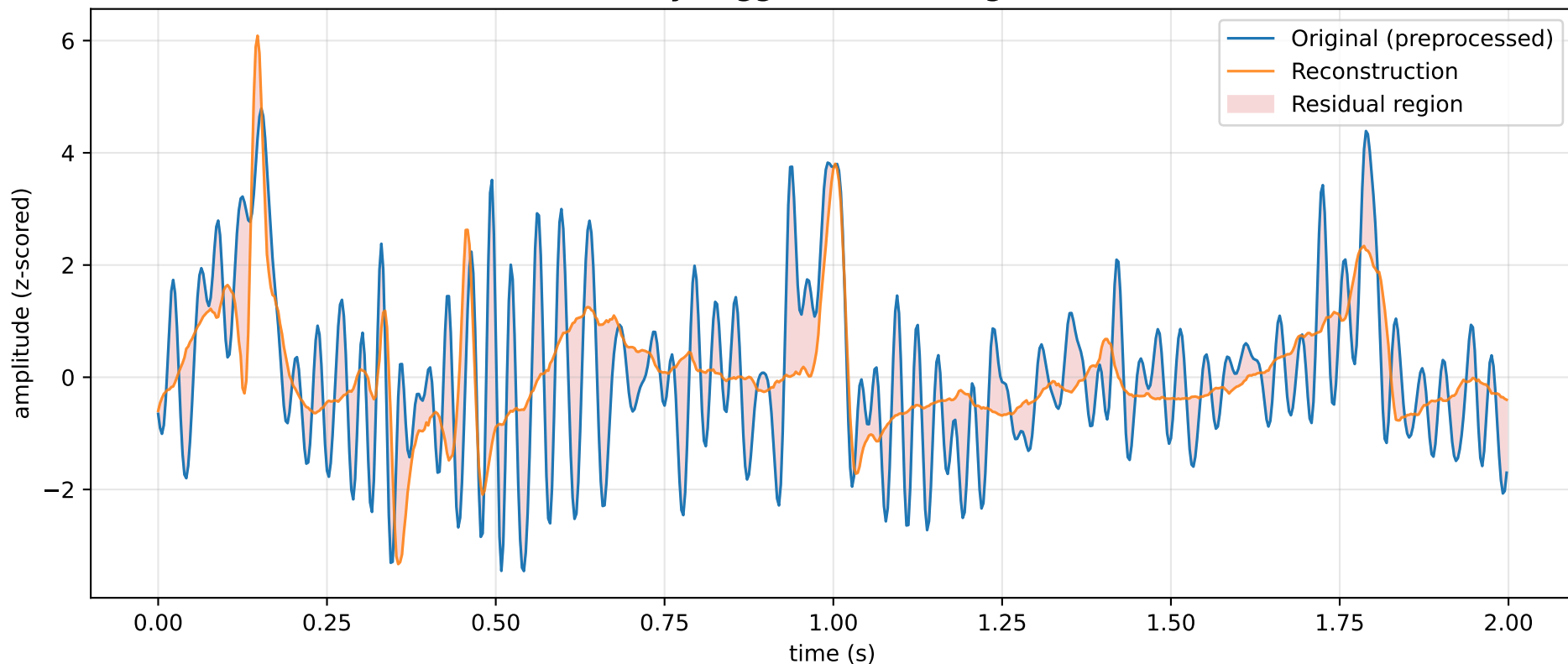
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 105, sample 444182
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 1.6444
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

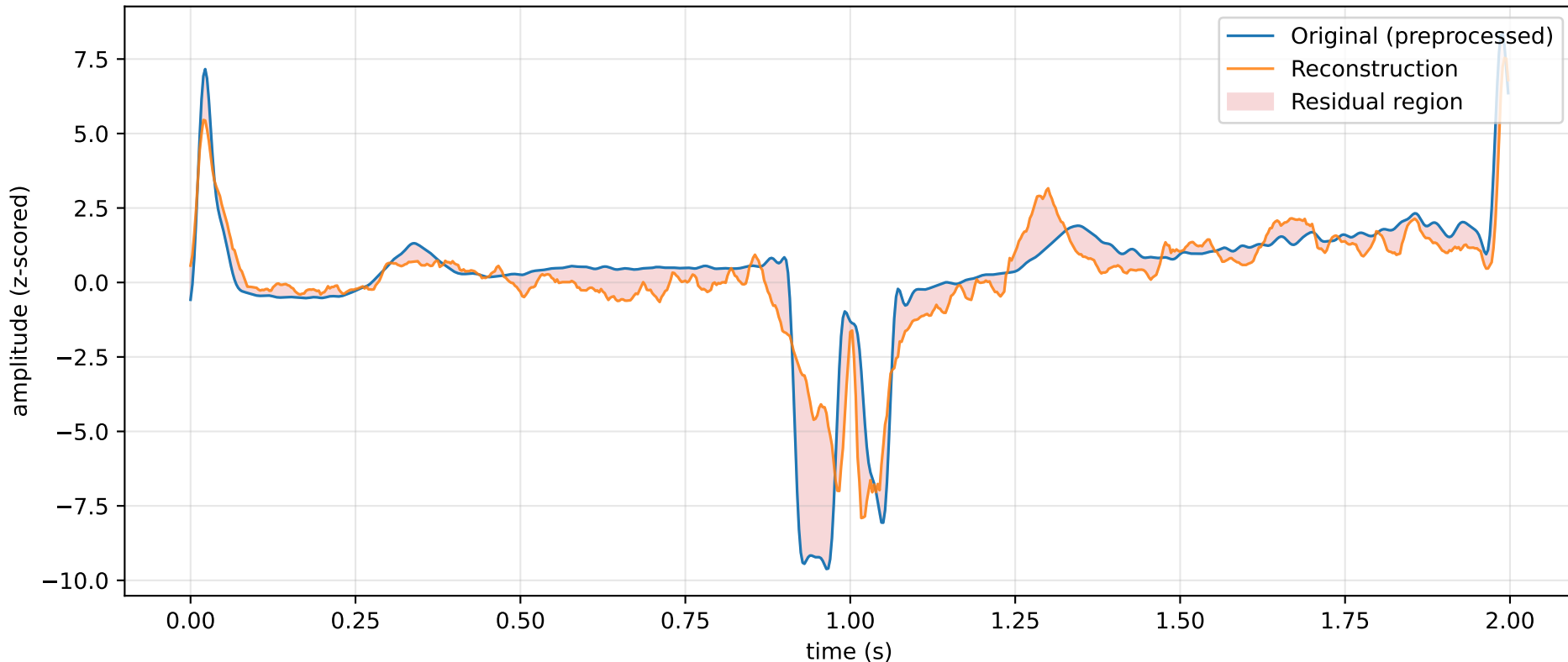
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115121
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 1.6326
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

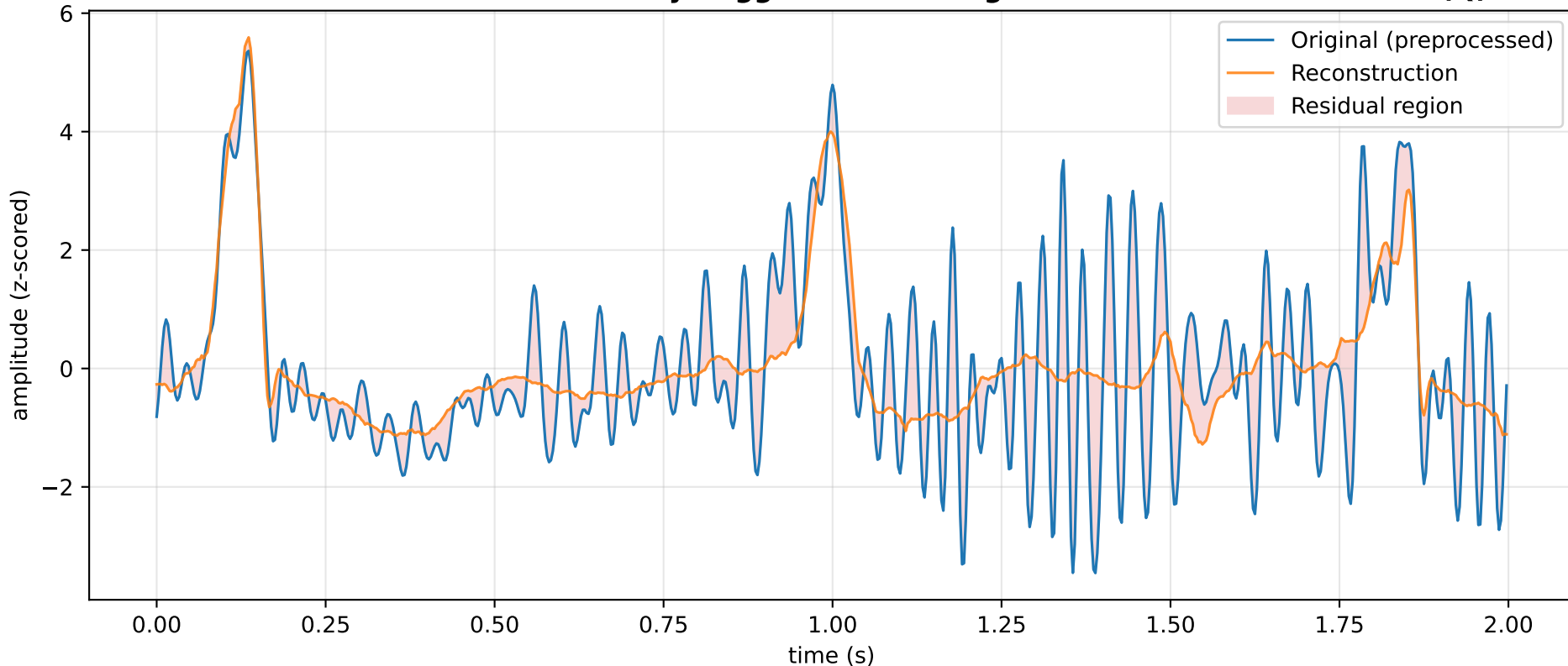
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 101, sample 114705
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 1.5325
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

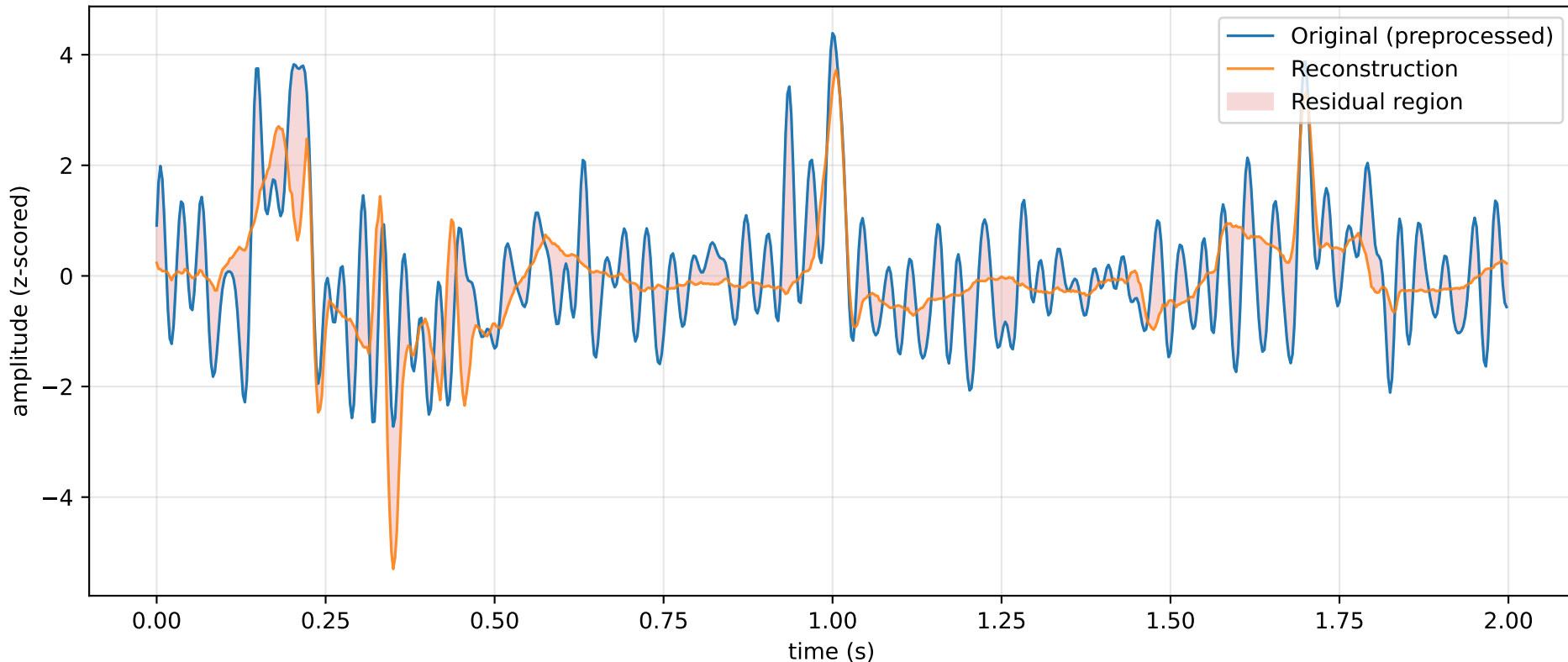
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 114816
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 1.2485
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

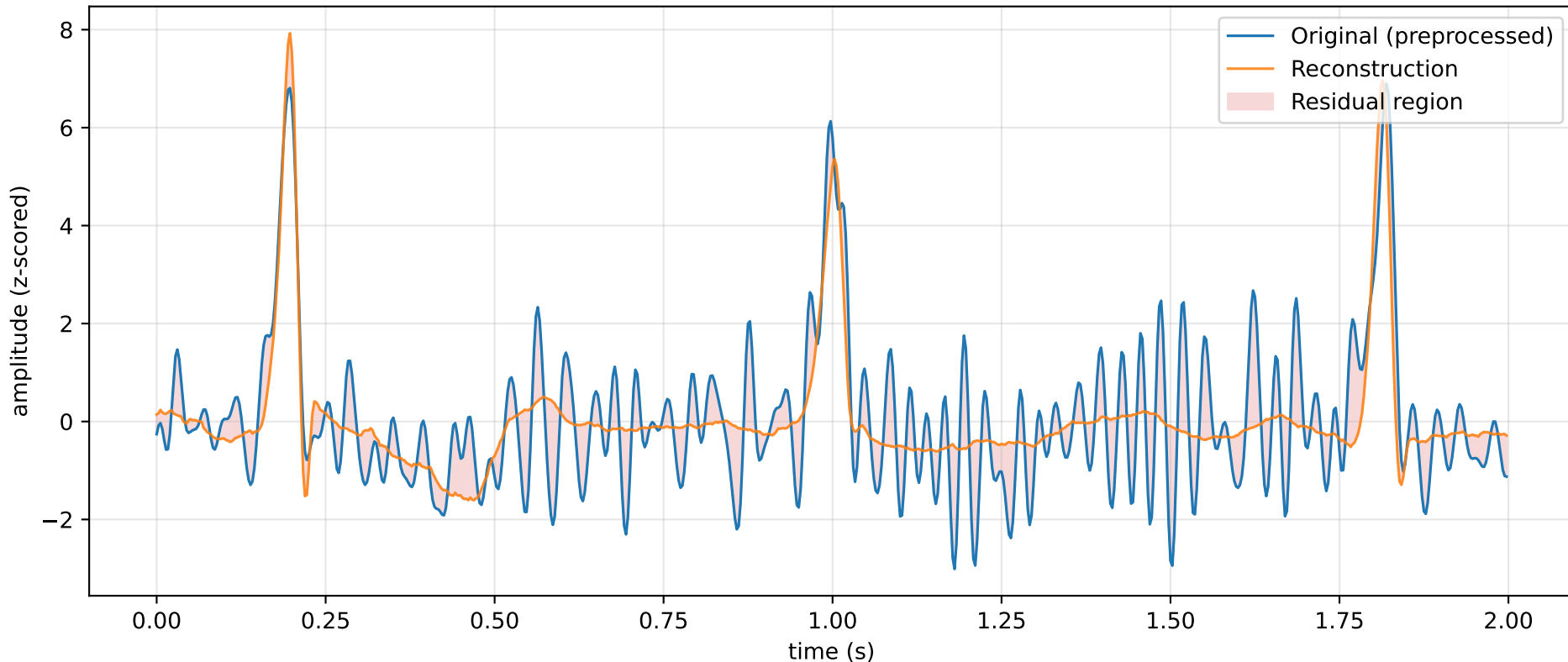
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 115405
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 1.0853
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

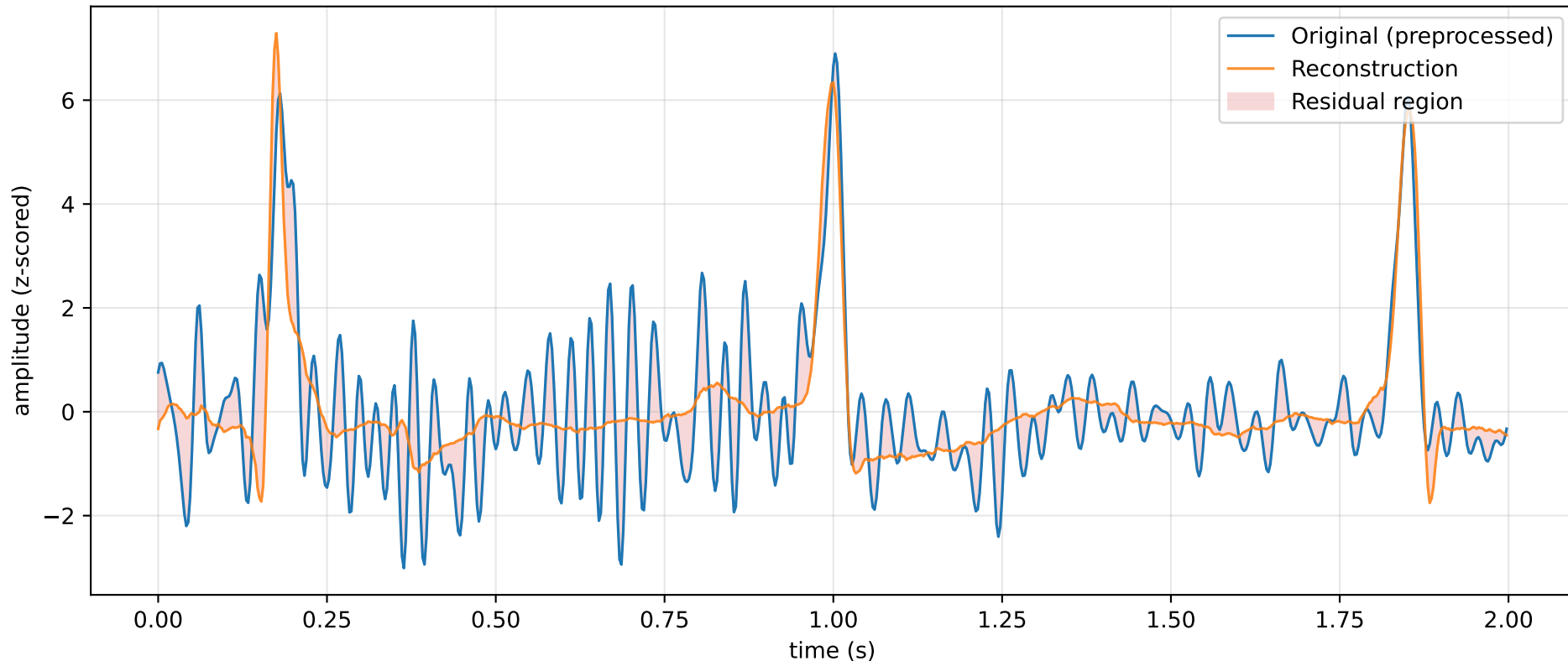
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 111856
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 1.0223
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

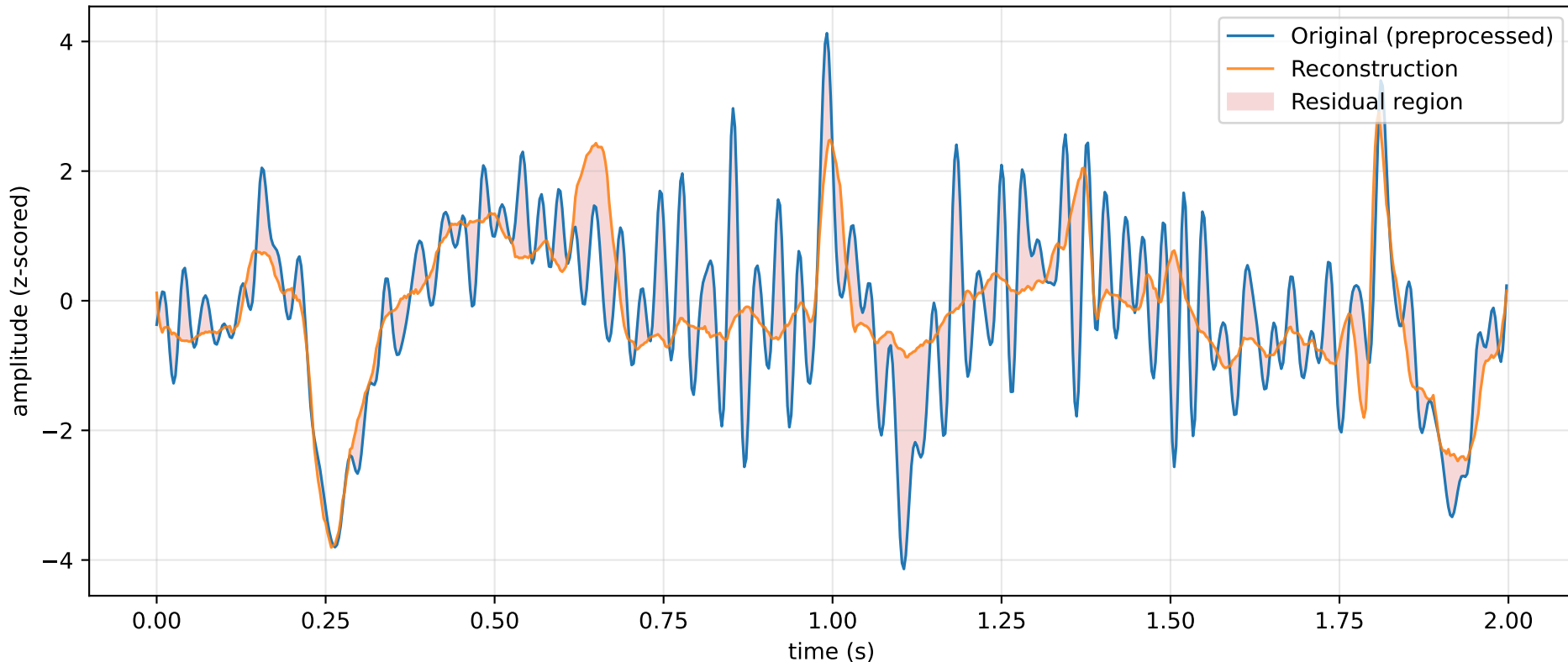
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 104, sample 112150
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 1.0054
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

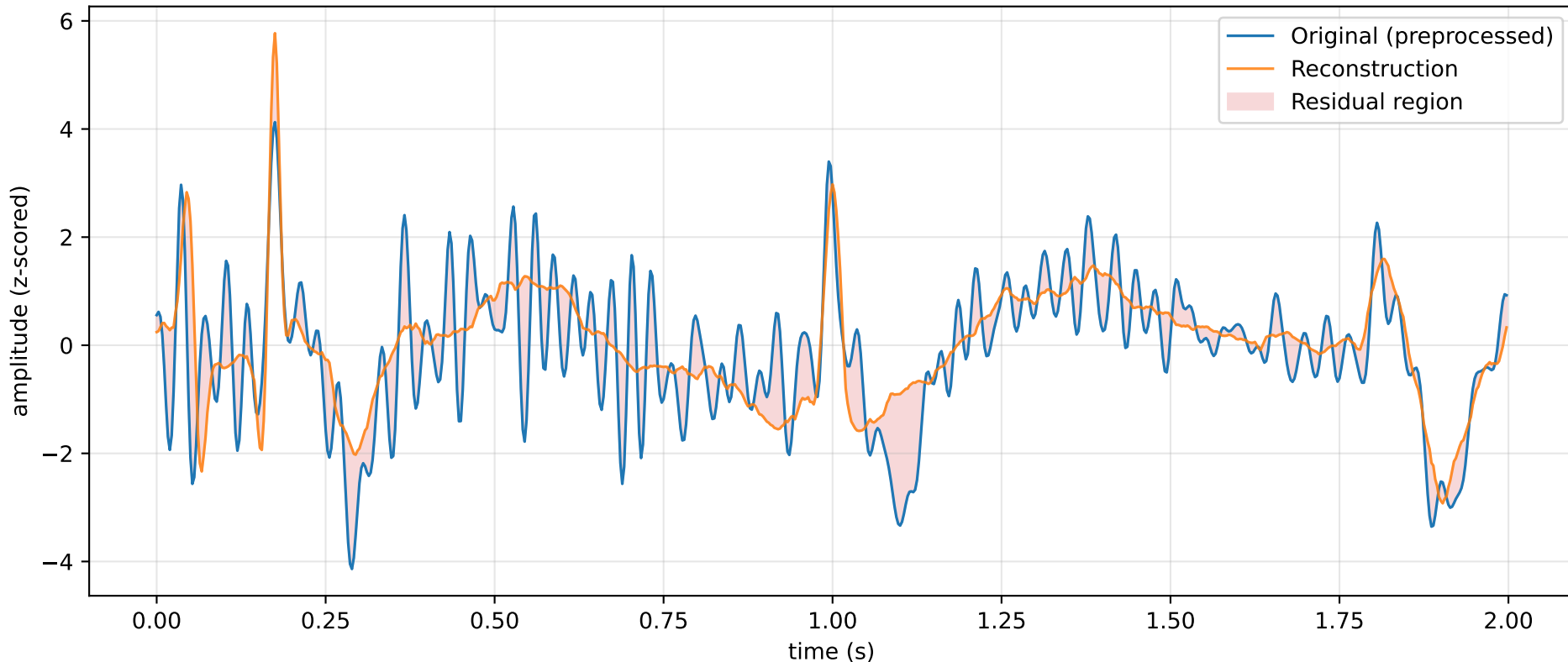
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Paced beat (/)



Source : rec 104, sample 47938
Beat symbol : / (Paced beat)
Ground truth : ANOMALY
Residual : 0.9589
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

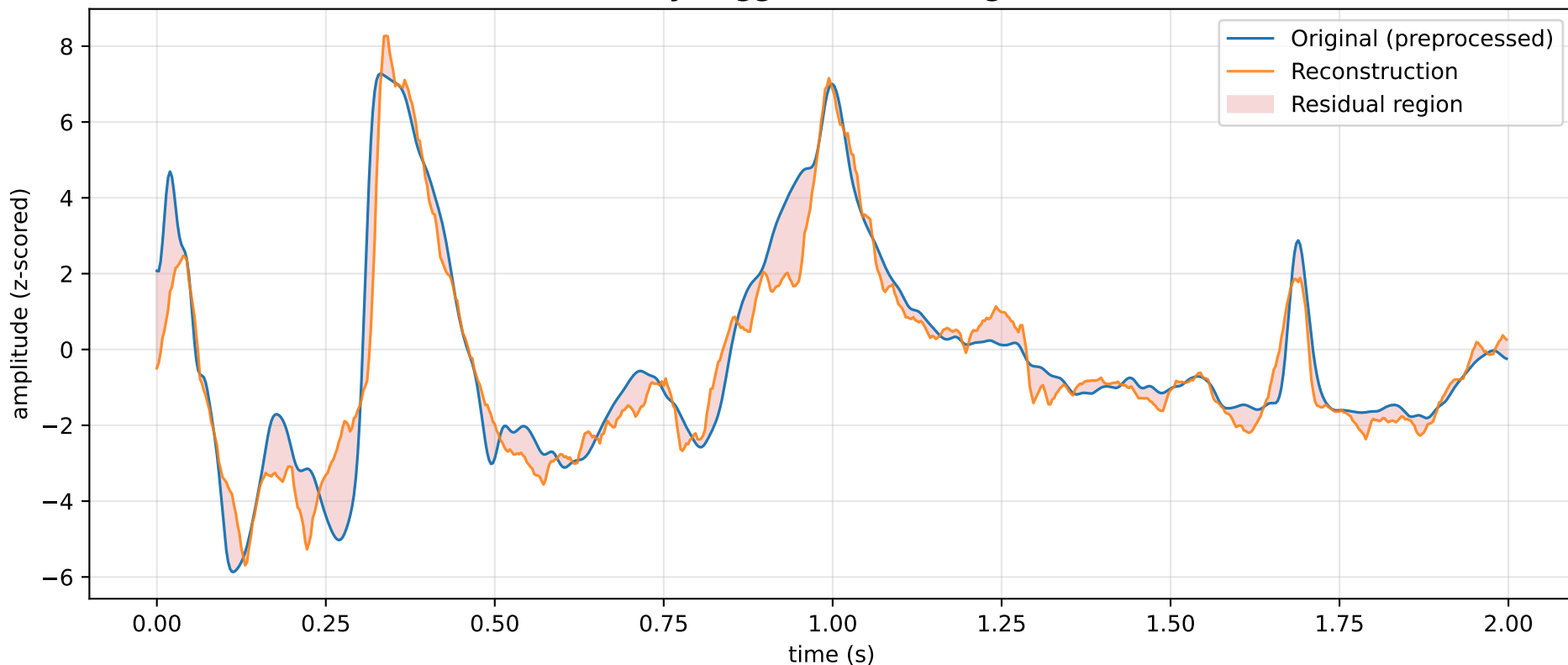
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Paced beat (/)



Source : rec 104, sample 48232
Beat symbol : / (Paced beat)
Ground truth : ANOMALY
Residual : 0.8561
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

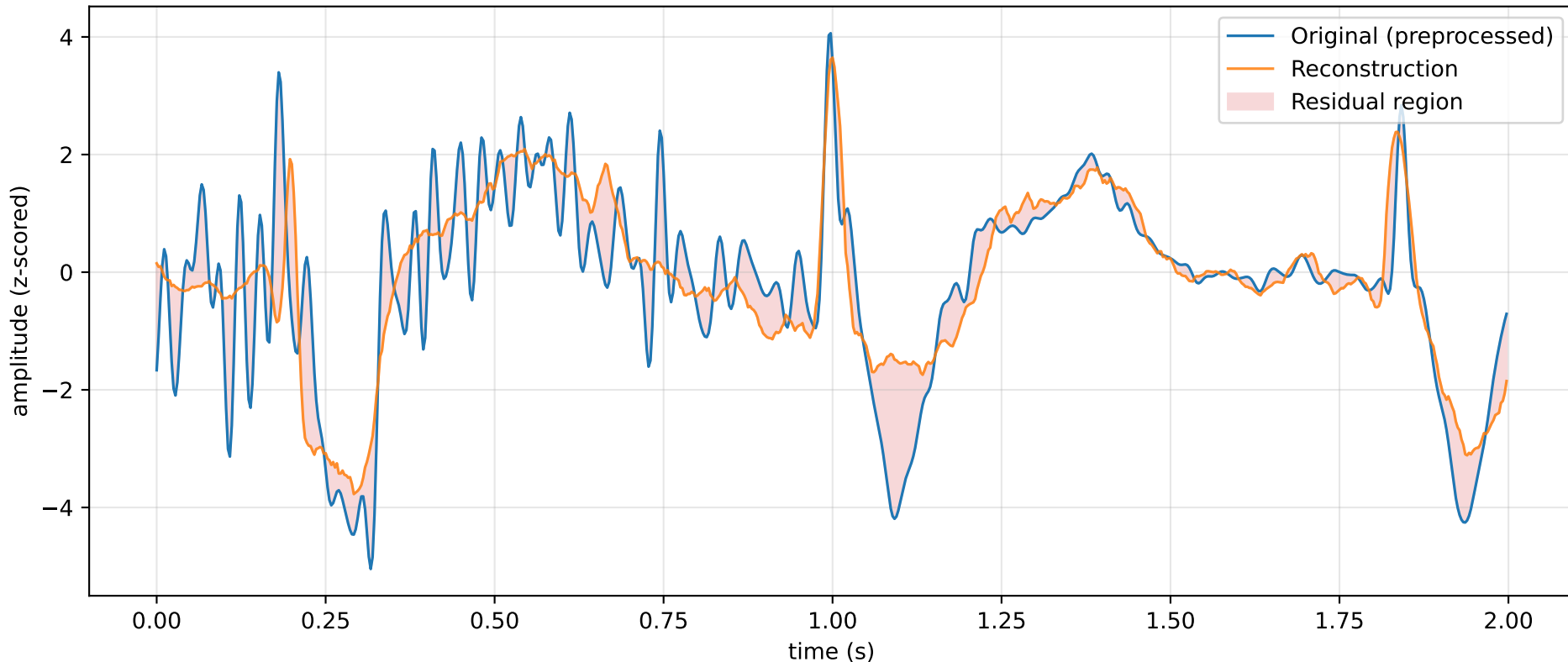
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



Source : rec 105, sample 595482
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 0.8405
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

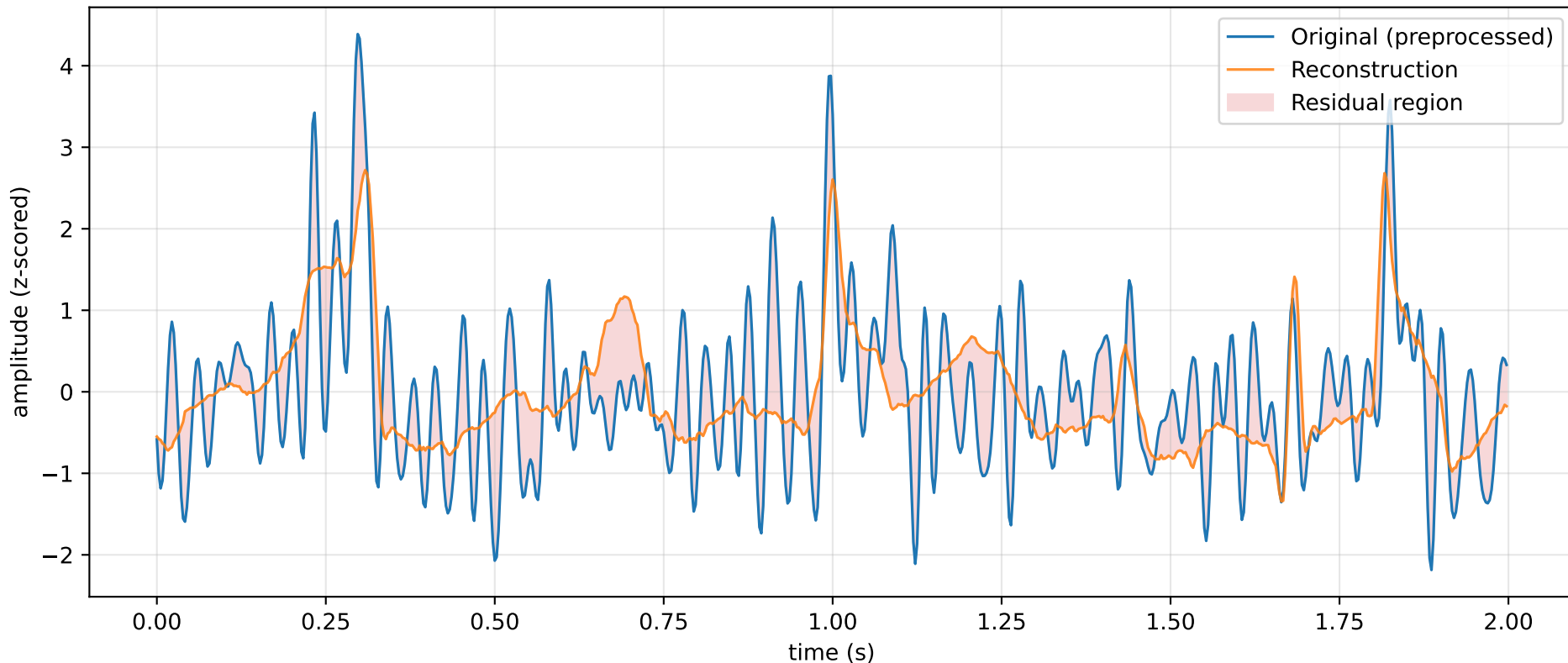
TRUE POSITIVE — model correctly flagged abnormal: ground truth = Paced beat (/)



Source : rec 104, sample 99248
Beat symbol : / (Paced beat)
Ground truth : ANOMALY
Residual : 0.8291
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

TRUE POSITIVE — model correctly flagged abnormal: ground truth = Unclassifiable (Q)



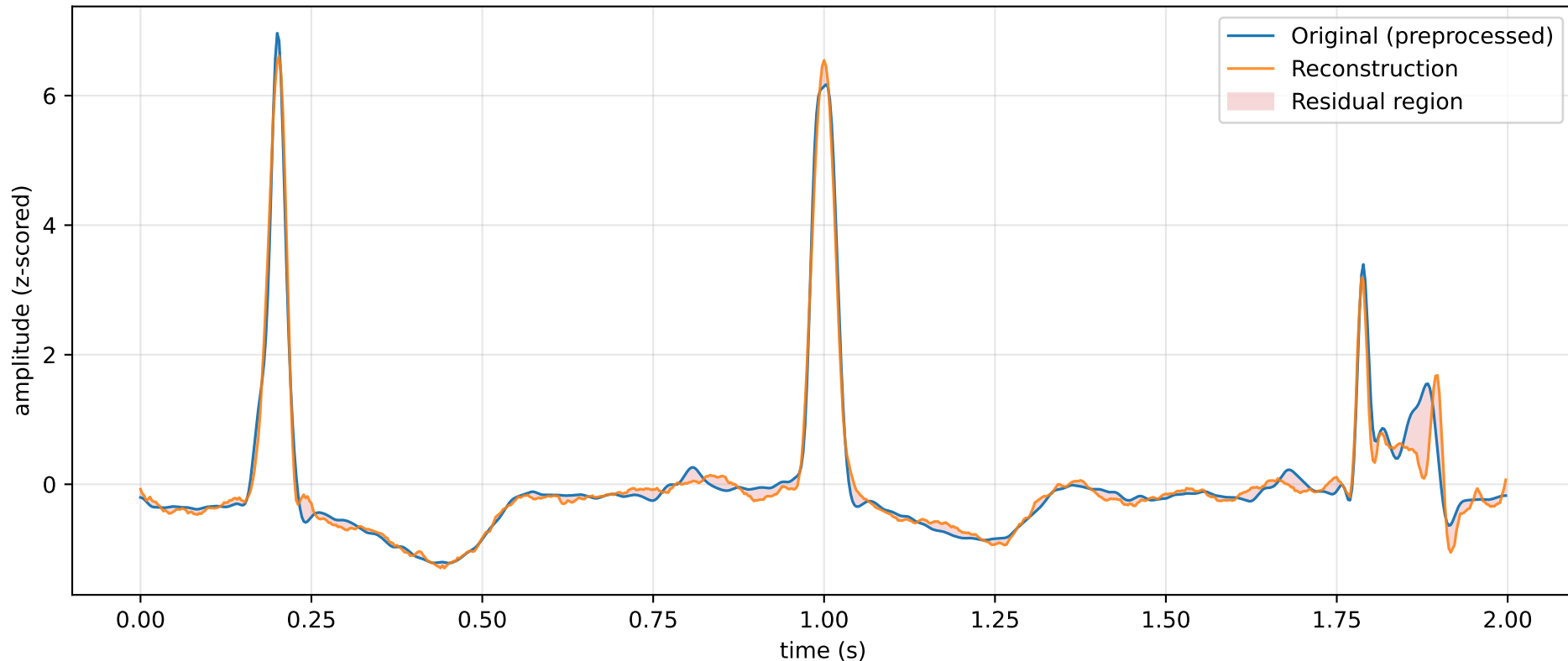
Source : rec 104, sample 115658
Beat symbol : Q (Unclassifiable)
Ground truth : ANOMALY
Residual : 0.7430
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

FALSE NEGATIVE — model MISSED an abnormal beat

(12 example(s) follow)

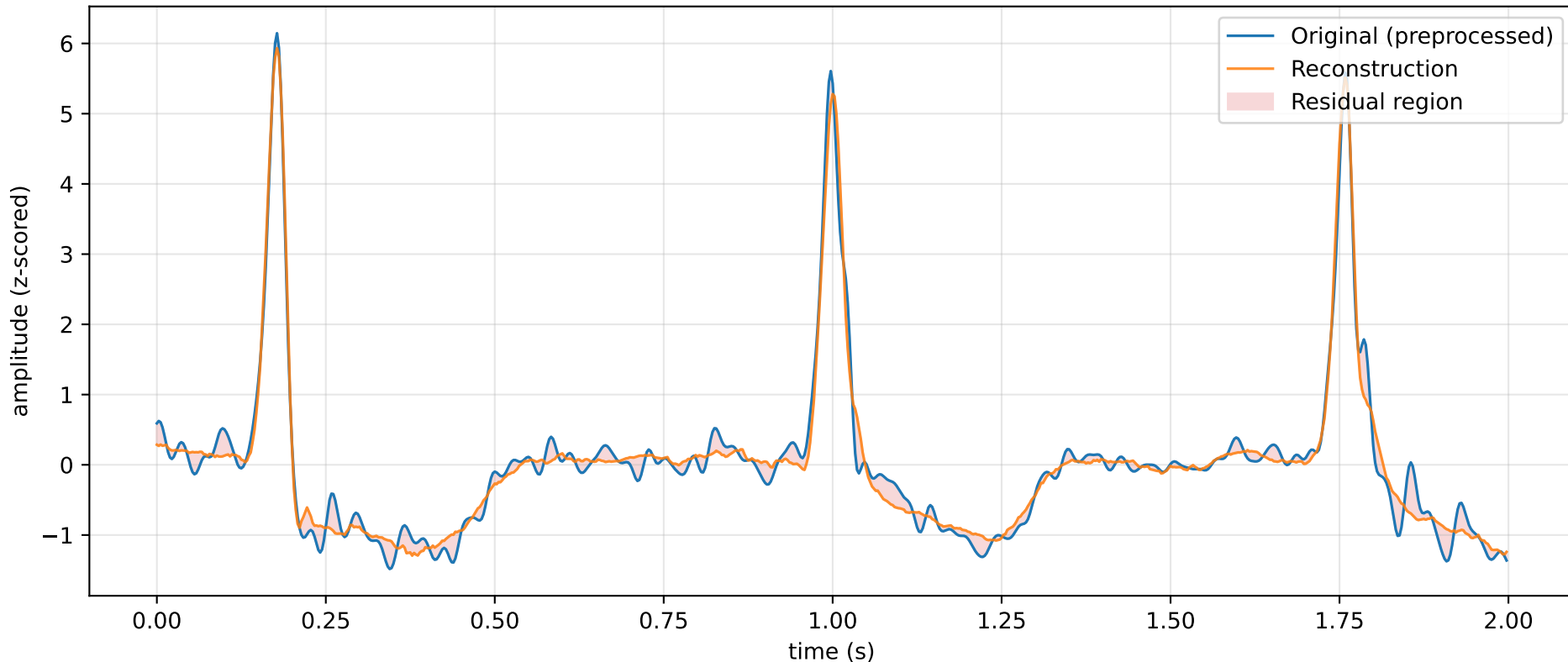
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 284556
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0433
Threshold : 0.0434
Model says : NORMAL

NORMAL

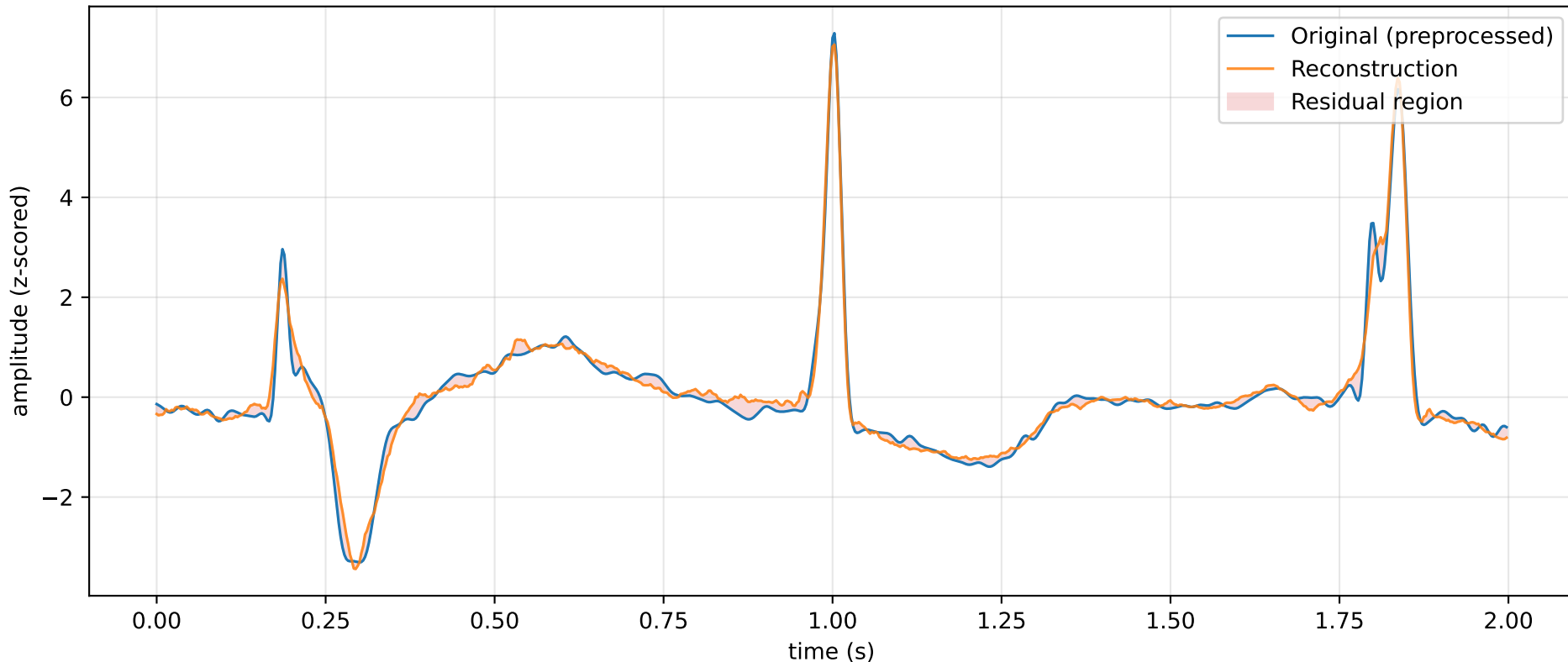
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 106625
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0433
Threshold : 0.0434
Model says : NORMAL

NORMAL

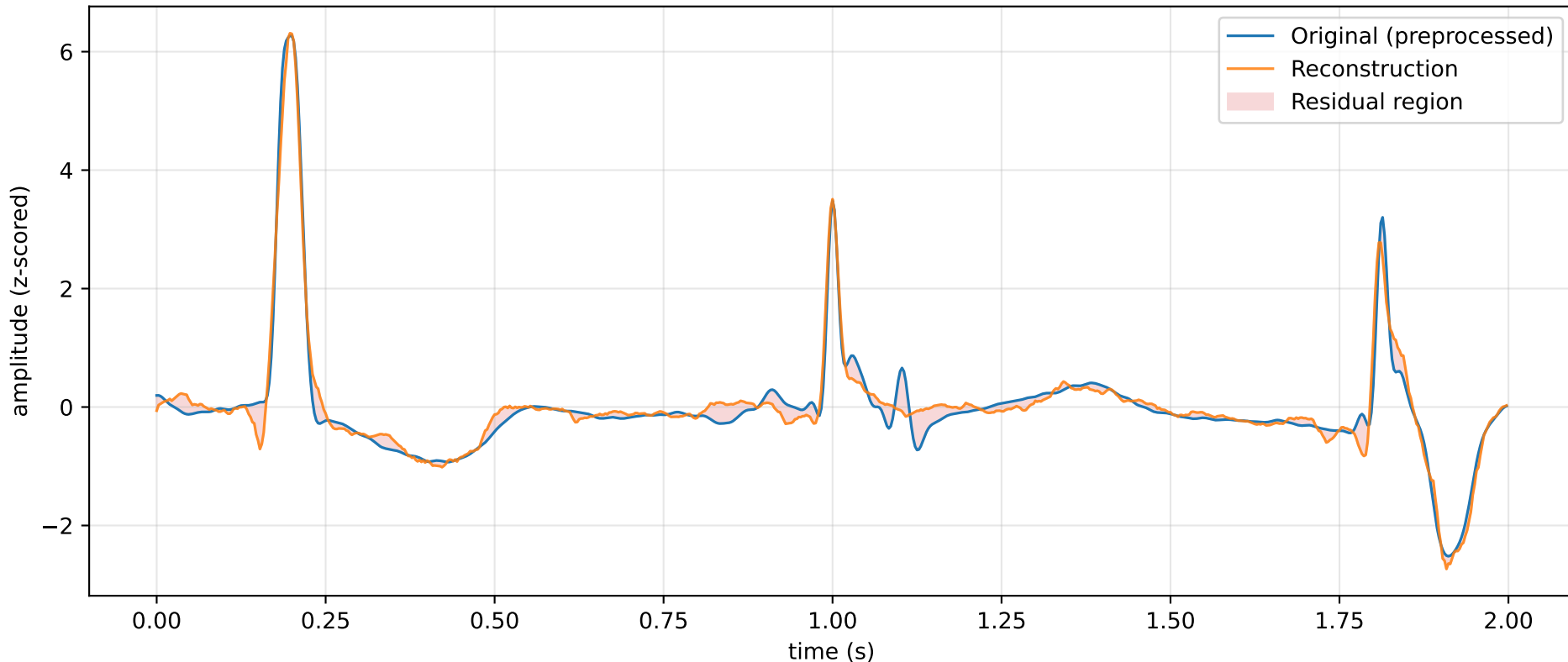
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 15816
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0432
Threshold : 0.0434
Model says : NORMAL

NORMAL

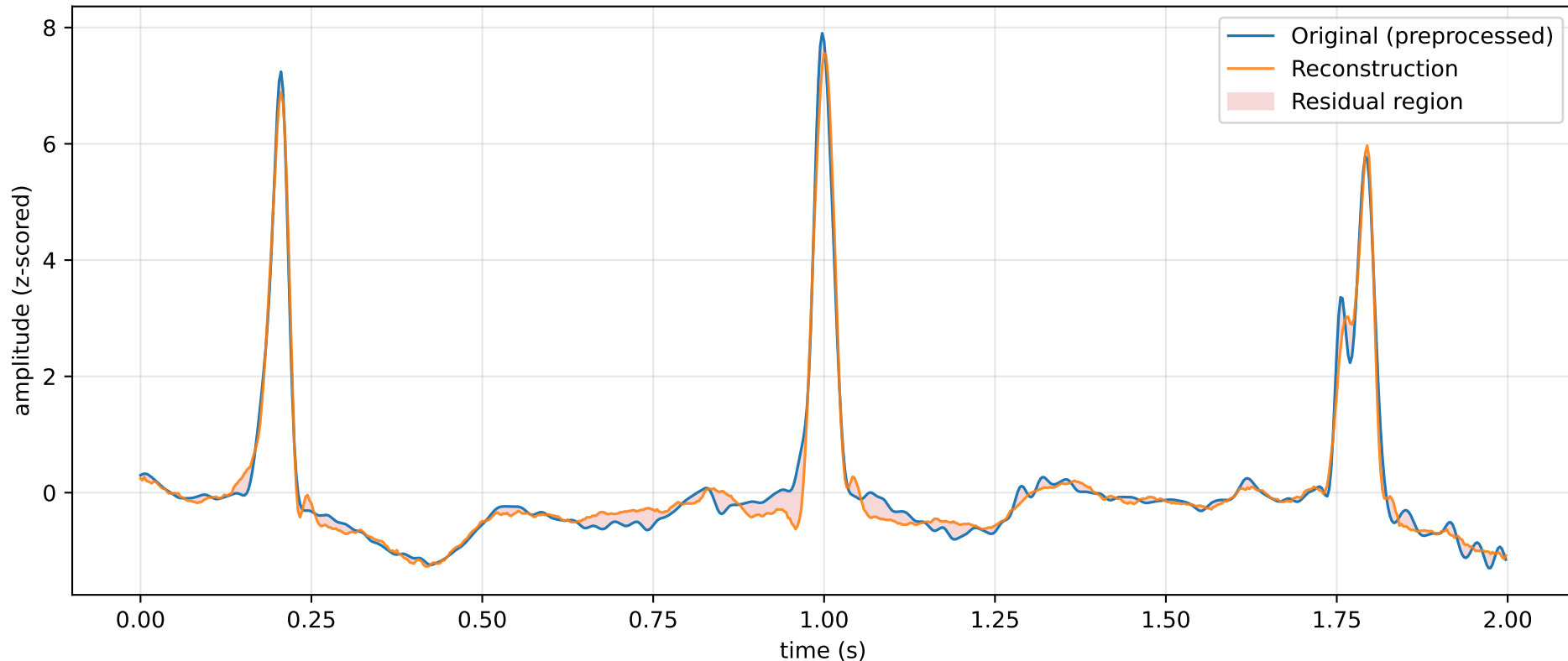
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 571806
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0432
Threshold : 0.0434
Model says : NORMAL

NORMAL

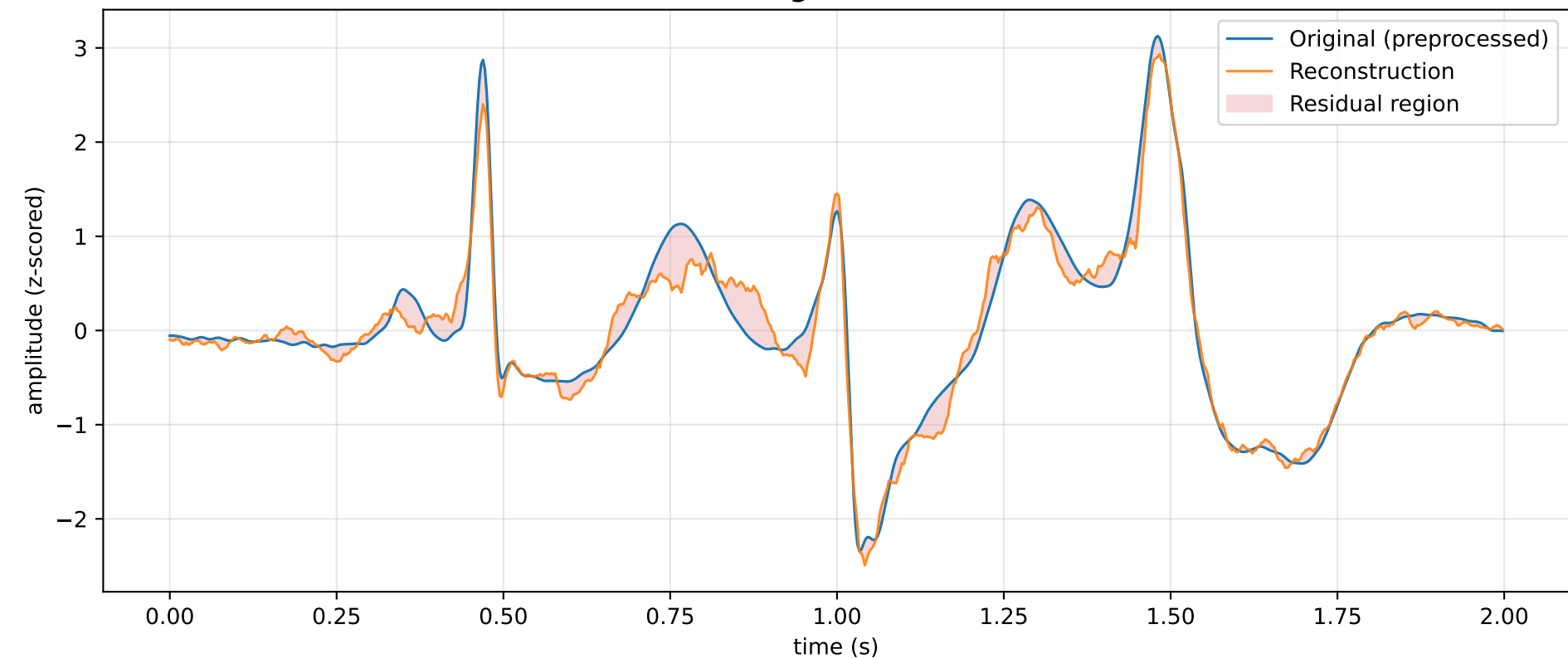
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 184873
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0430
Threshold : 0.0434
Model says : NORMAL

NORMAL

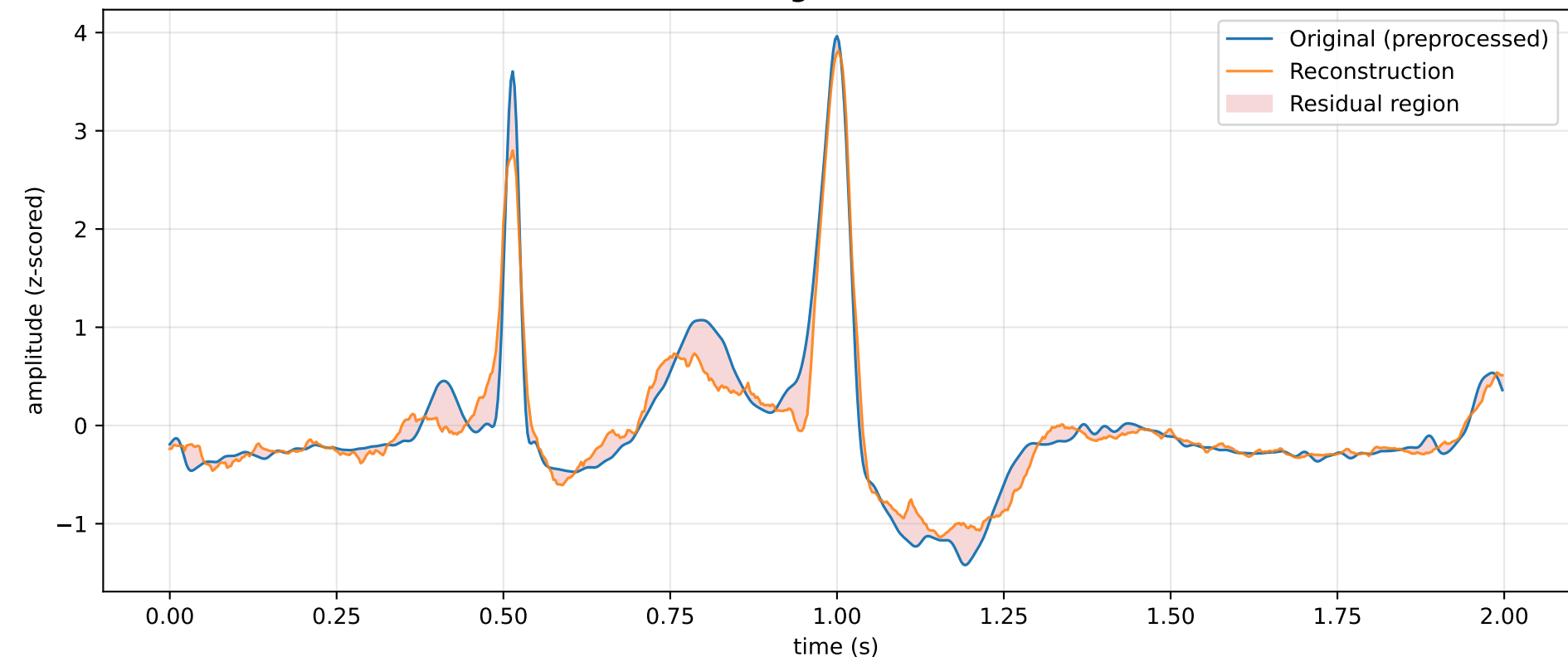
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Premature ventricular contraction (PVC)



Source : rec 106, sample 611355
Beat symbol : V (Premature ventricular contraction (PVC))
Ground truth : ANOMALY
Residual : 0.0428
Threshold : 0.0434
Model says : NORMAL

NORMAL

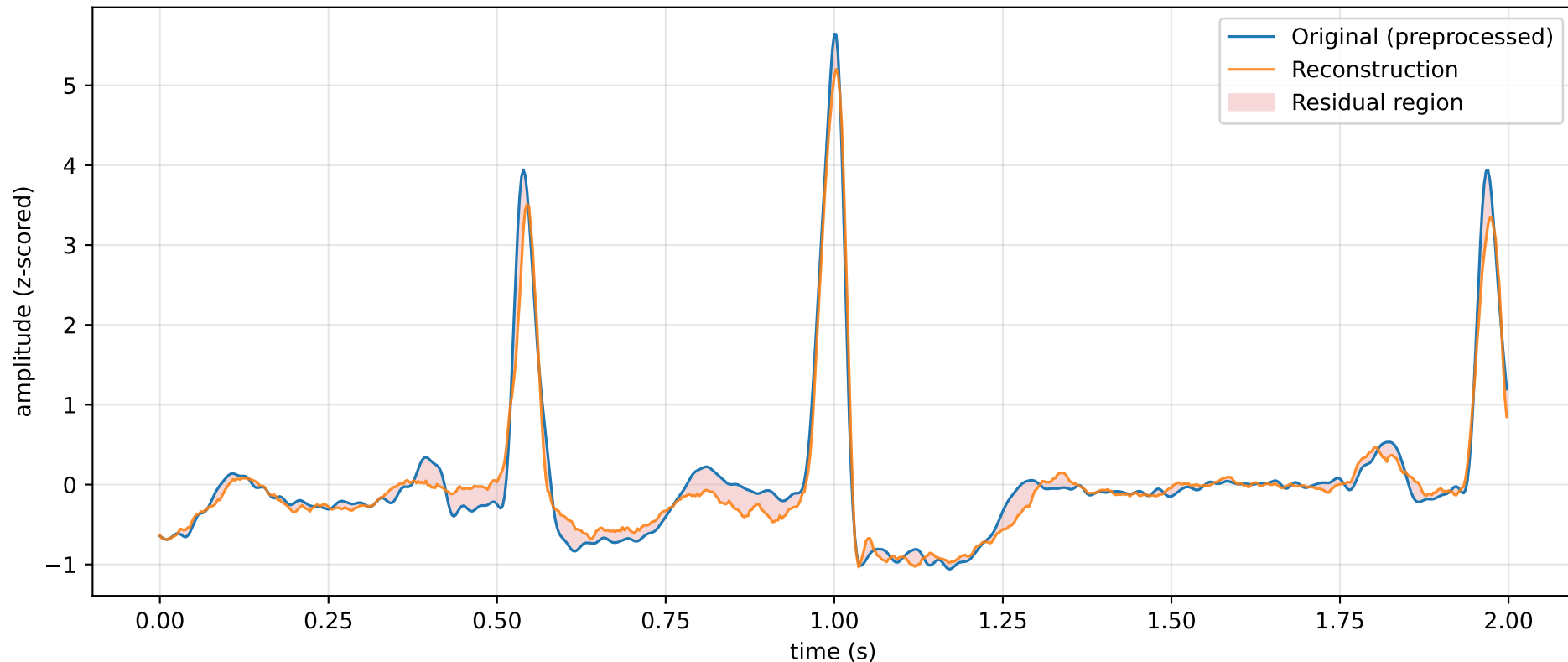
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Premature ventricular contraction (PVC)



Source : rec 106, sample 534246
Beat symbol : V (Premature ventricular contraction (PVC))
Ground truth : ANOMALY
Residual : 0.0427
Threshold : 0.0434
Model says : NORMAL

NORMAL

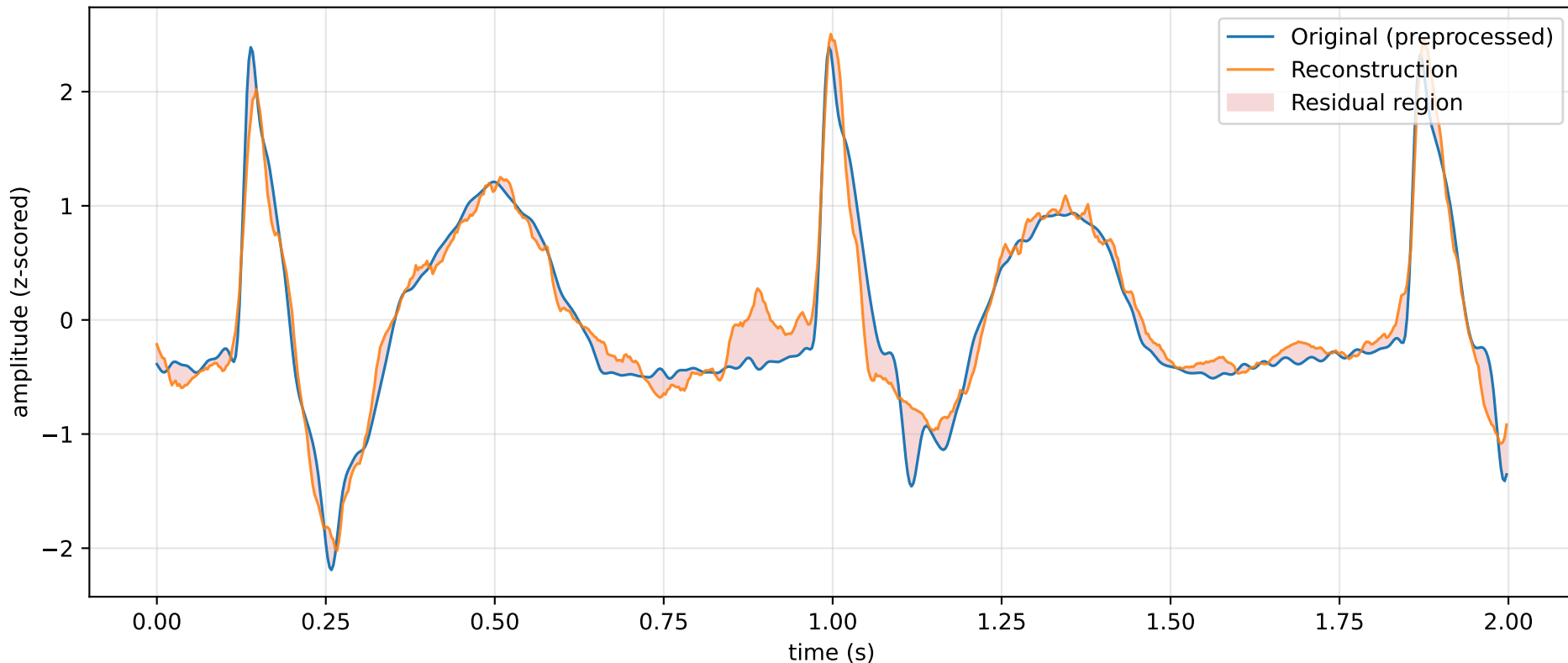
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Premature ventricular contraction (PVC)



Source : rec 105, sample 24842
Beat symbol : V (Premature ventricular contraction (PVC))
Ground truth : ANOMALY
Residual : 0.0425
Threshold : 0.0434
Model says : NORMAL

NORMAL

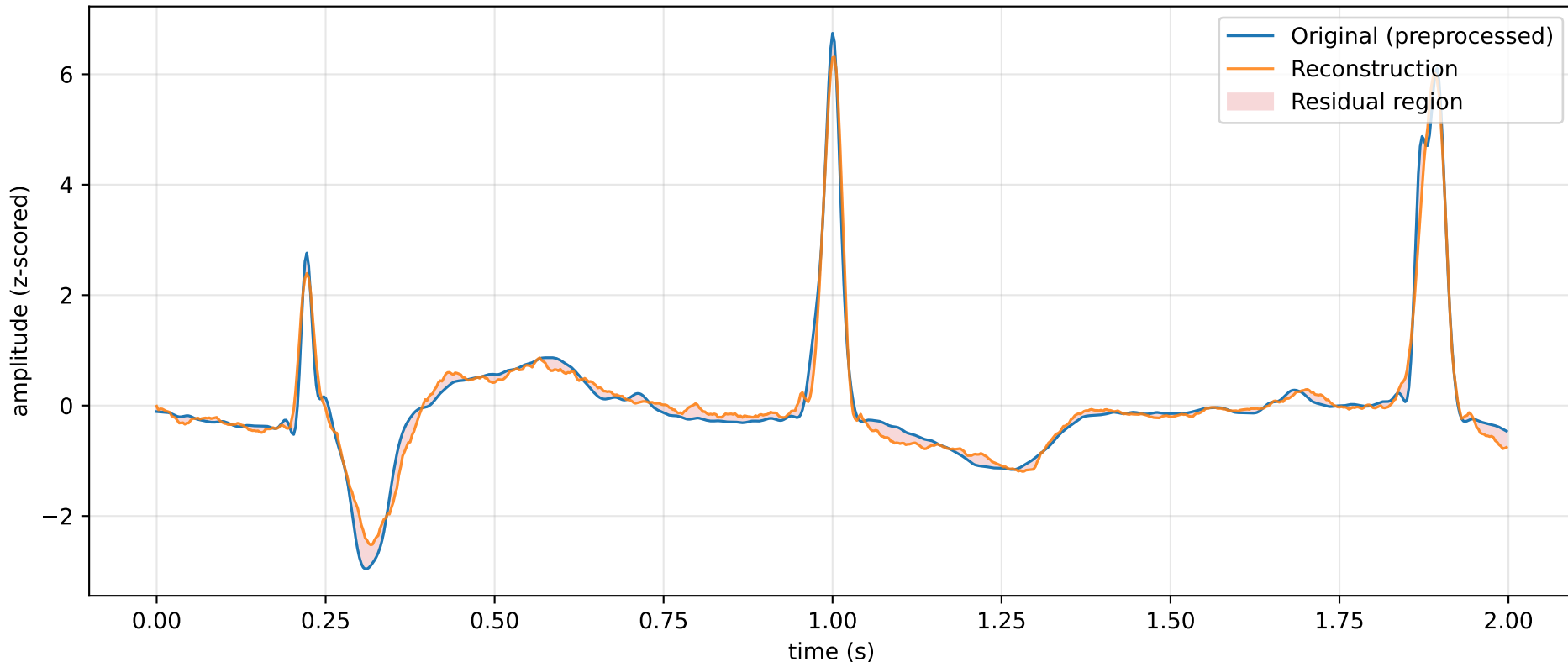
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Paced beat (/)



Source : rec 107, sample 158559
Beat symbol : / (Paced beat)
Ground truth : ANOMALY
Residual : 0.0424
Threshold : 0.0434
Model says : NORMAL

NORMAL

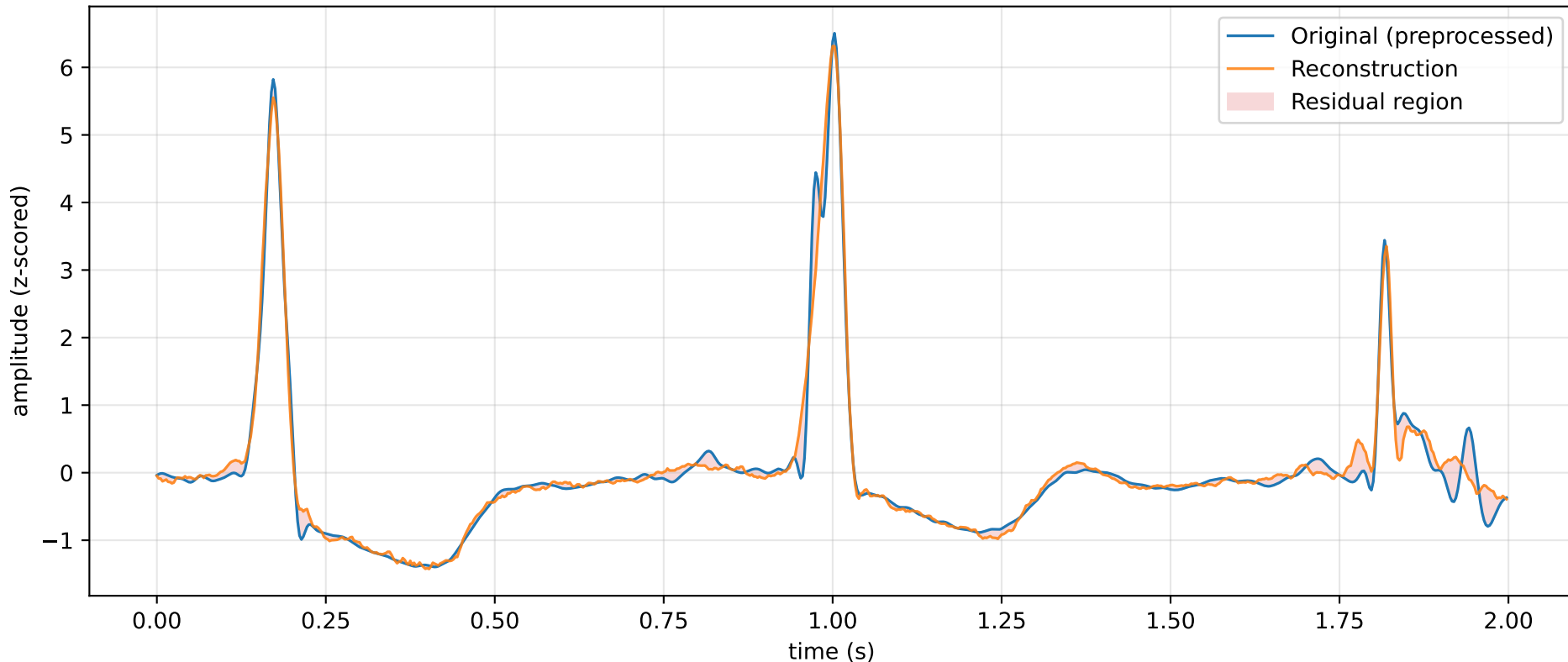
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 597801
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0421
Threshold : 0.0434
Model says : NORMAL

NORMAL

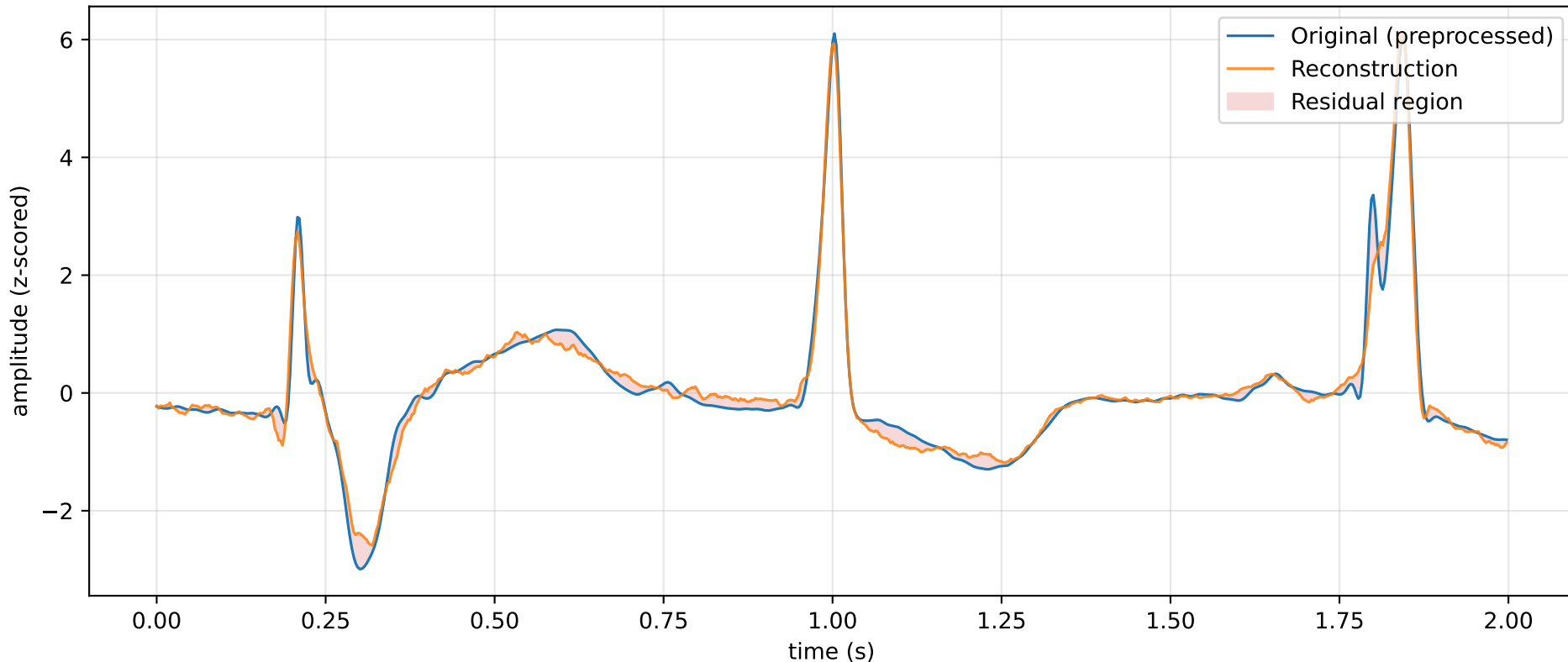
FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



Source : rec 104, sample 299730
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0421
Threshold : 0.0434
Model says : NORMAL

NORMAL

FALSE NEGATIVE — model MISSED an abnormal beat: ground truth = Fusion of paced and normal (f)



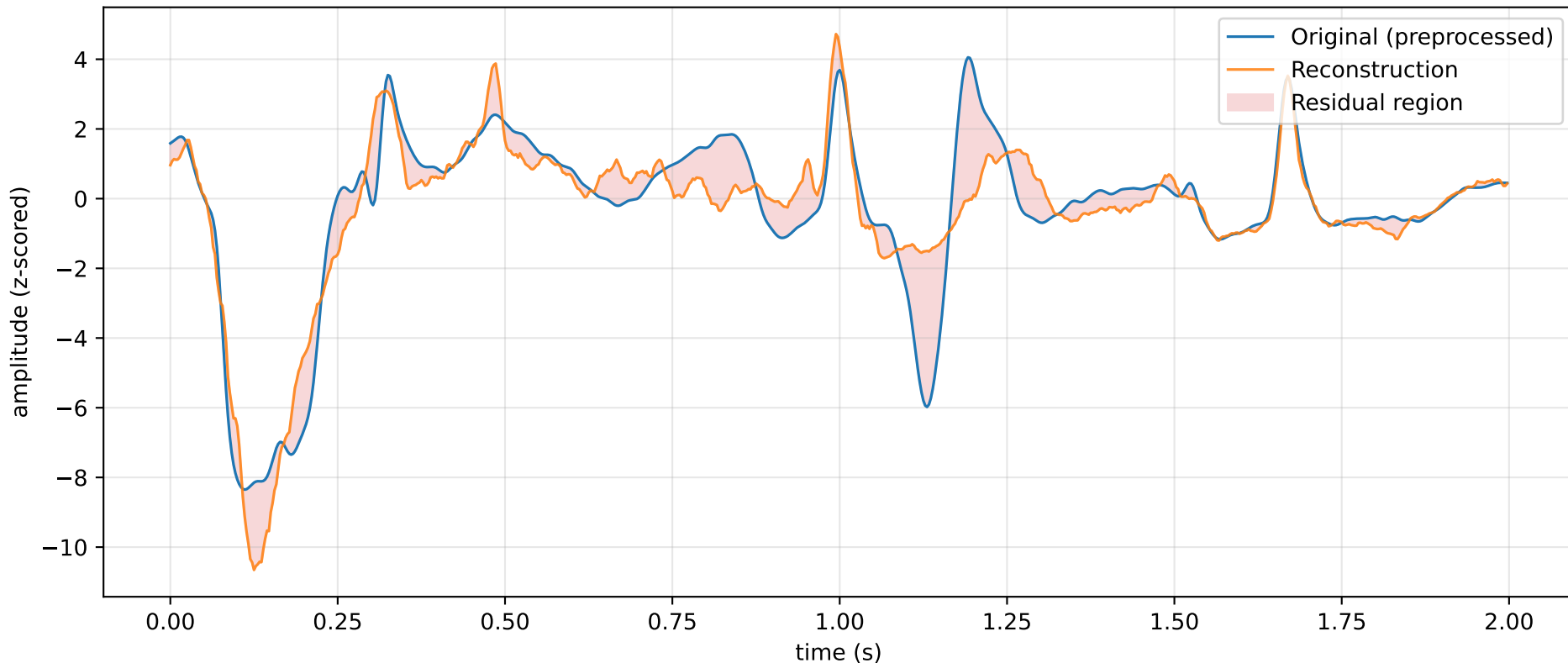
Source : rec 104, sample 592849
Beat symbol : f (Fusion of paced and normal)
Ground truth : ANOMALY
Residual : 0.0420
Threshold : 0.0434
Model says : NORMAL

NORMAL

FALSE POSITIVE — model wrongly flagged a normal beat

(12 example(s) follow)

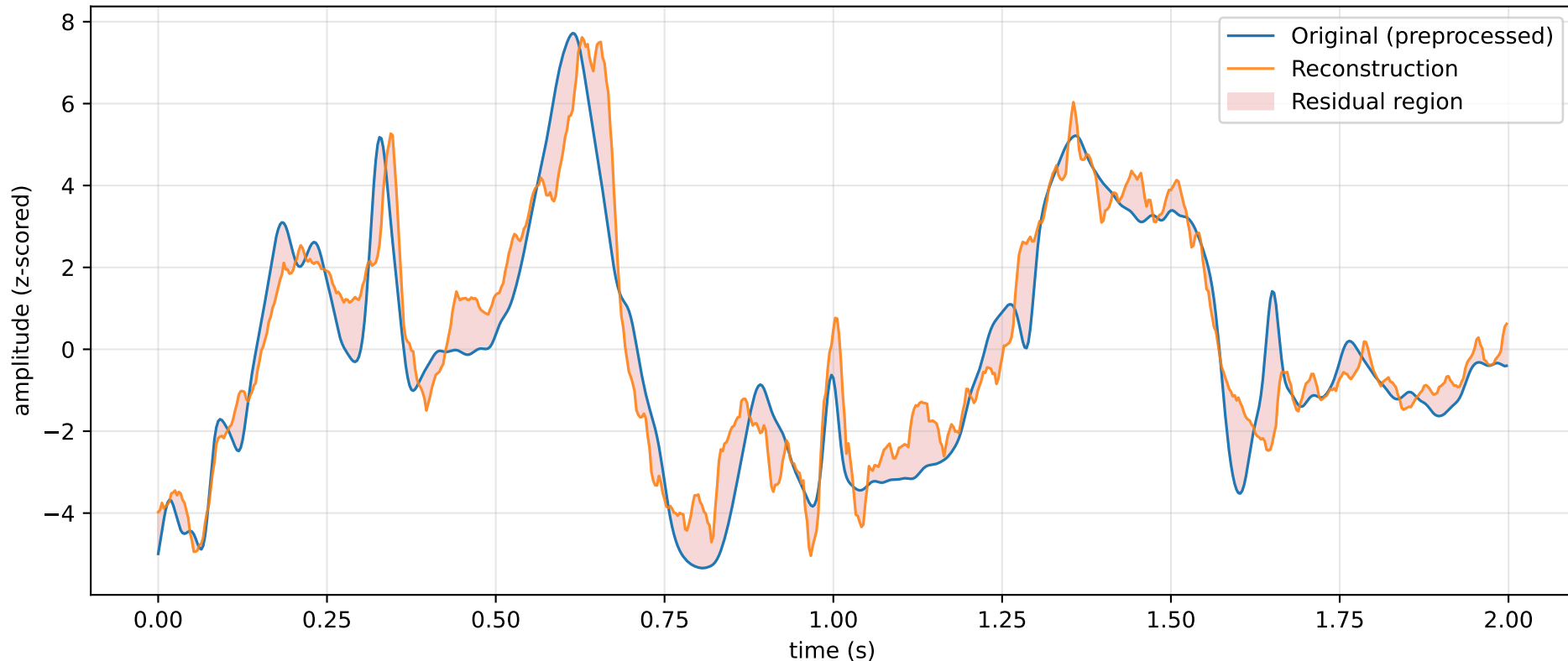
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 457427
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 1.1841
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

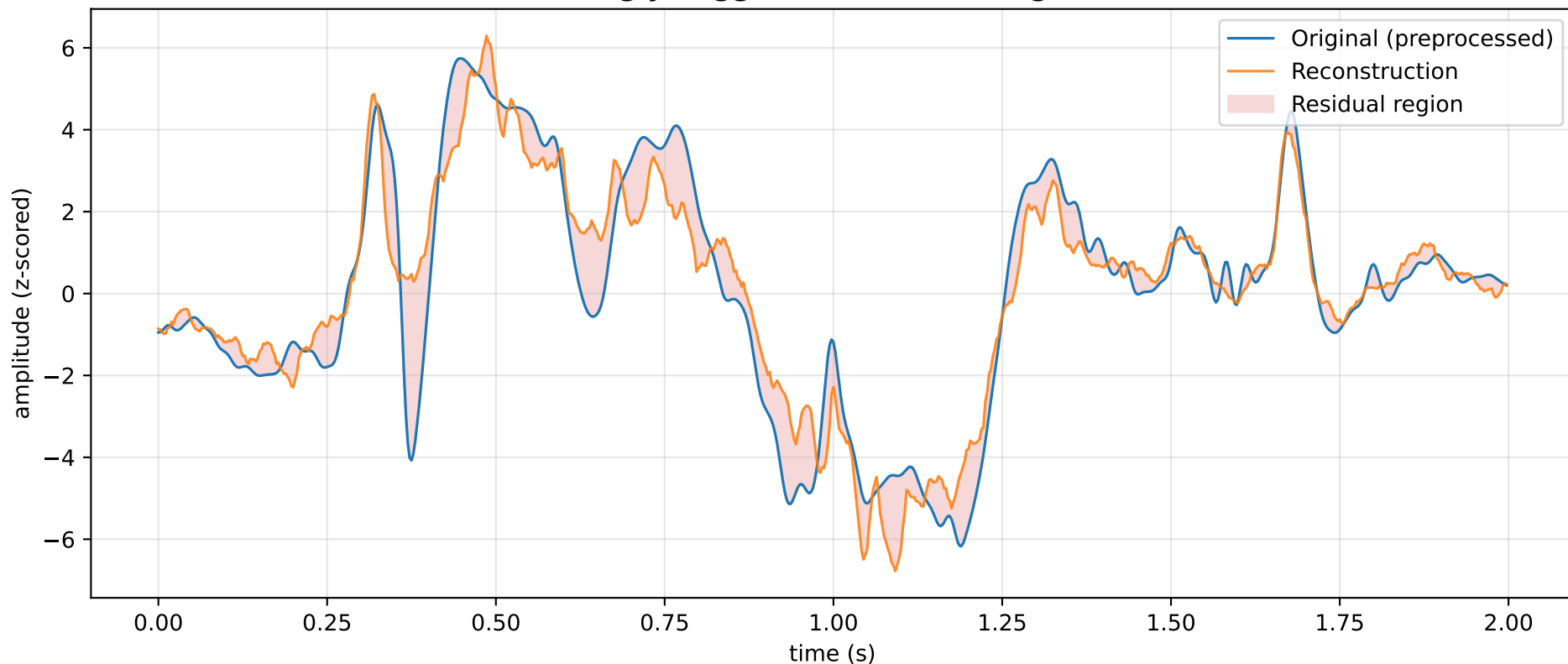
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 477791
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 1.1486
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

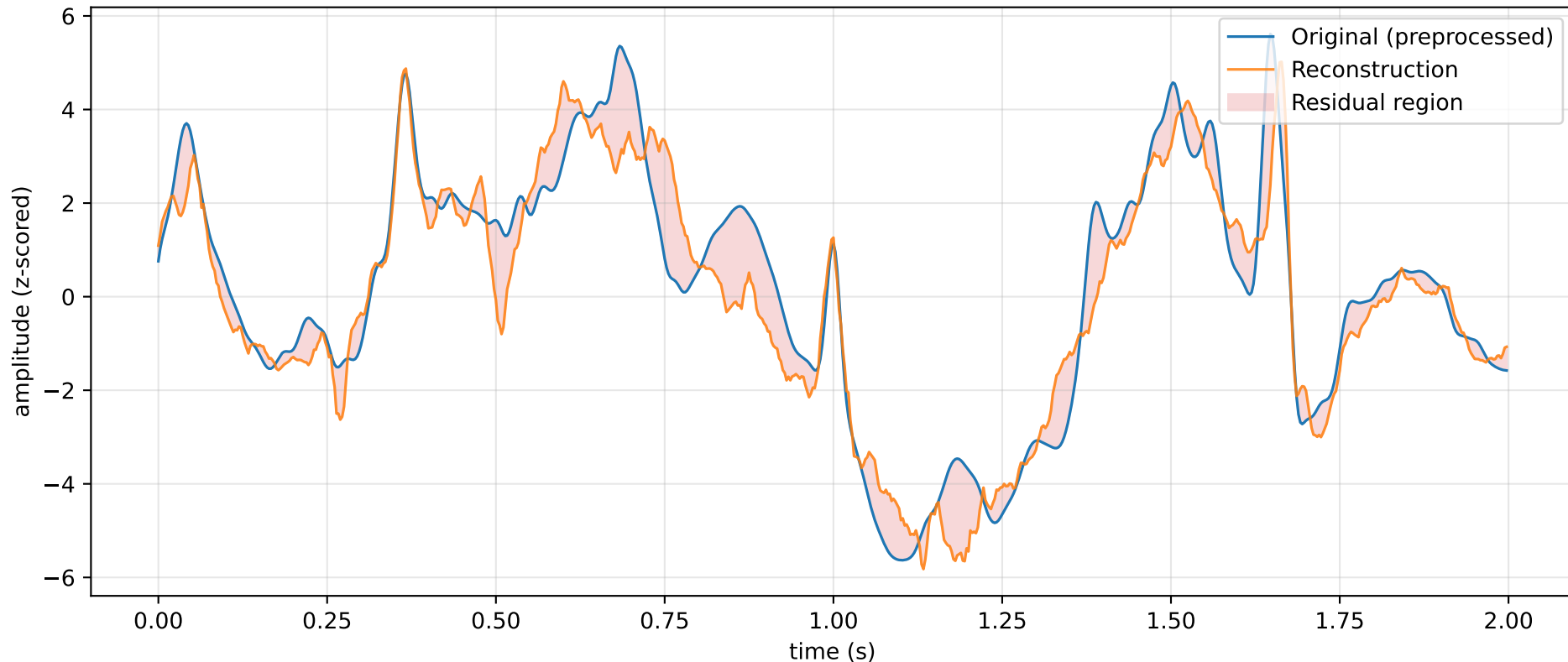
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 435895
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 1.0606
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

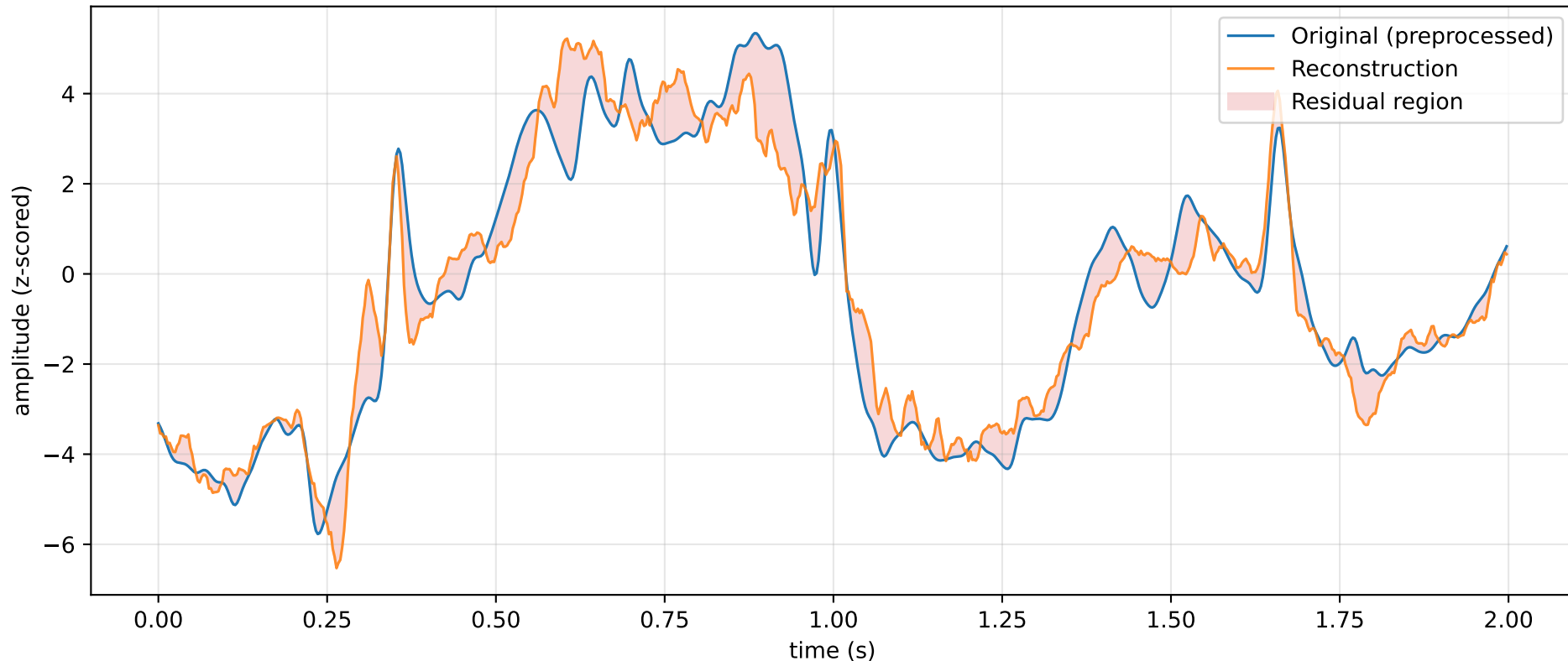
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 441811
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.8911
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

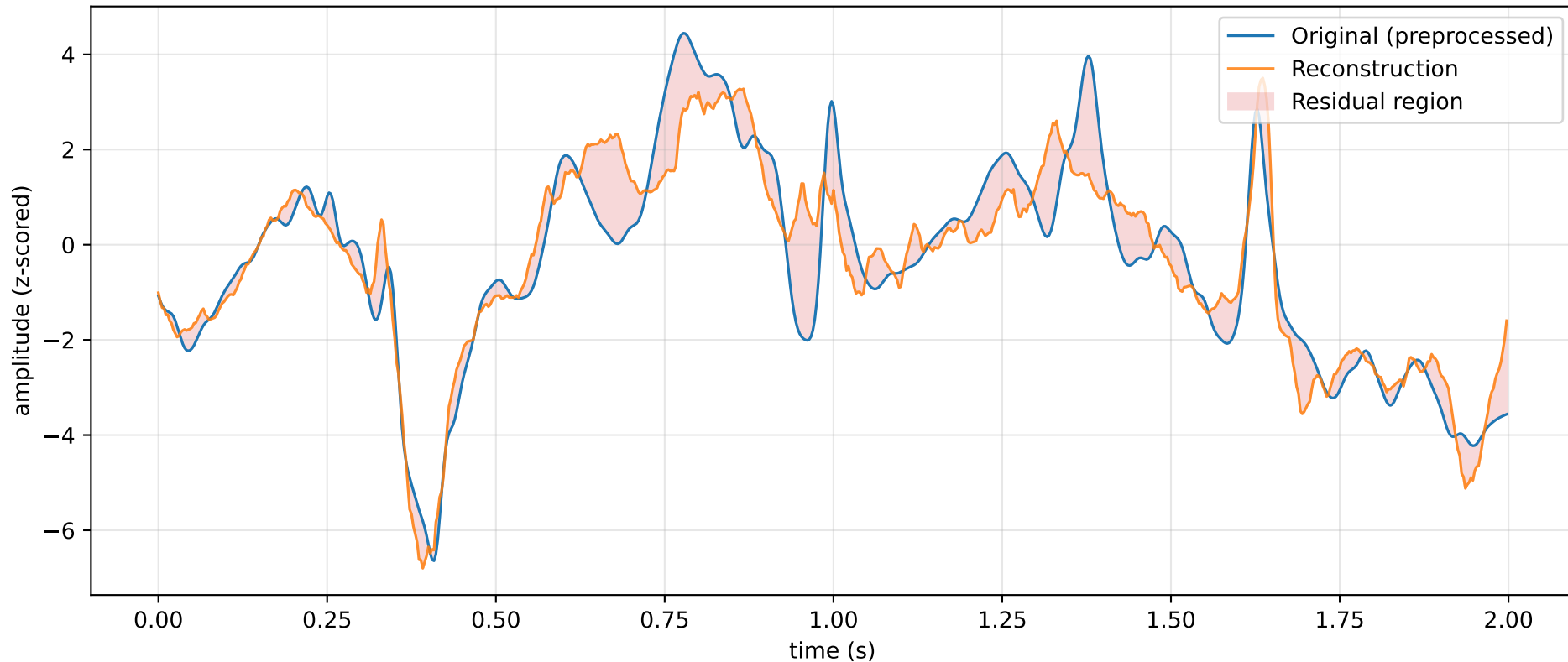
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 430628
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.8842
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

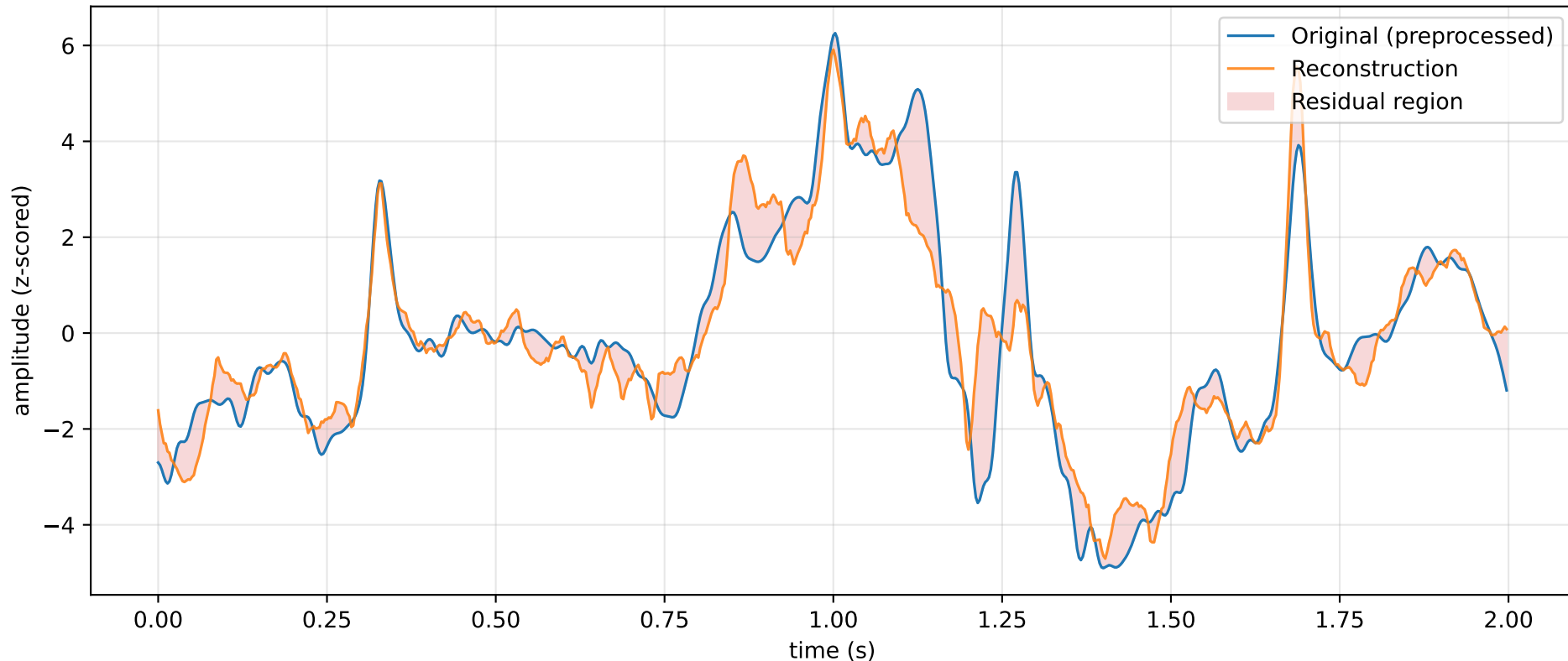
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 439248
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.7986
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

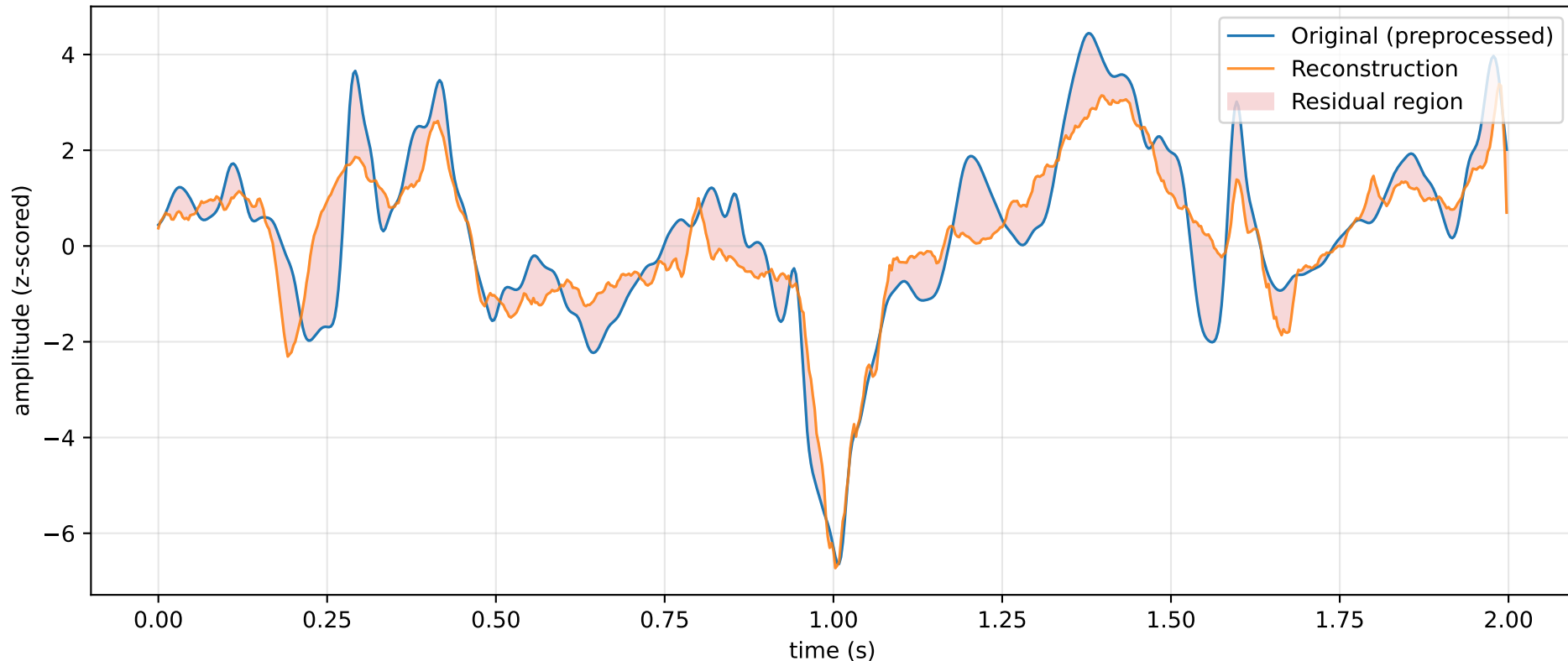
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 479189
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.7792
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

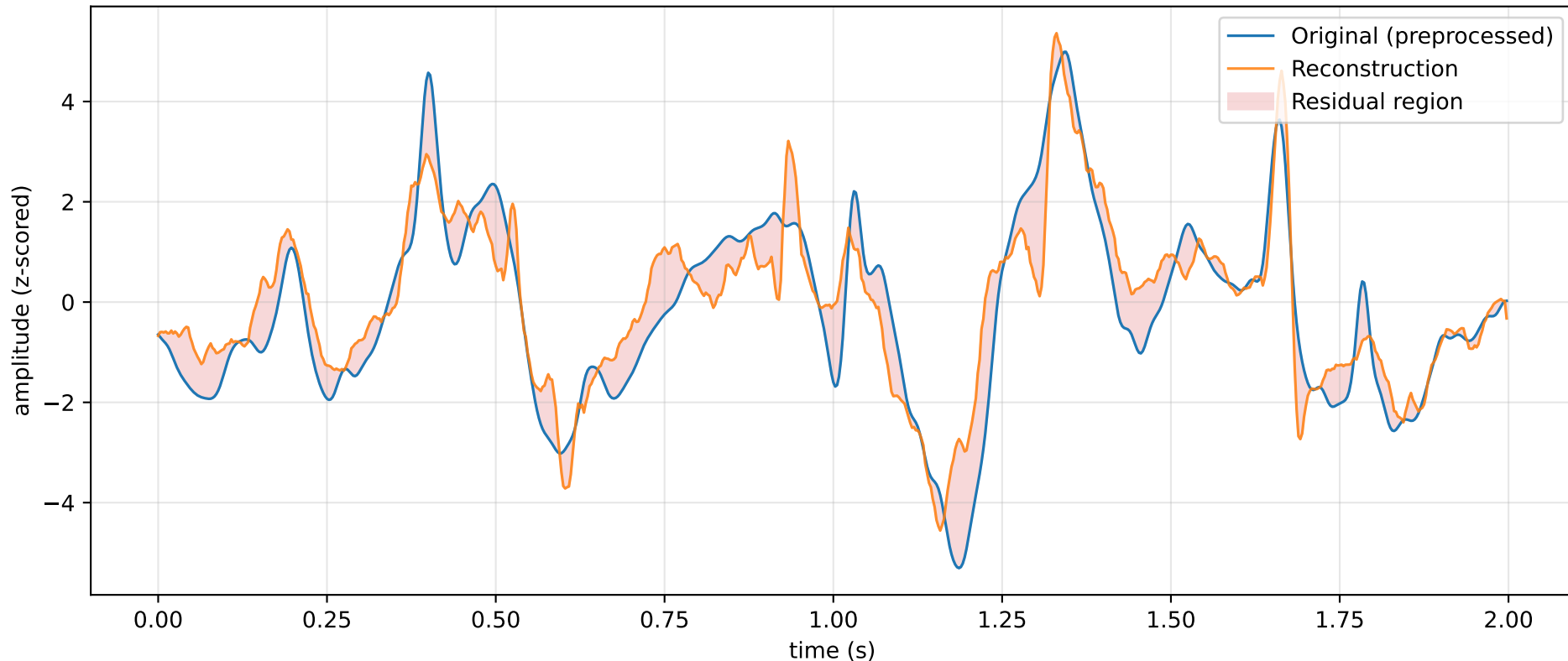
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 439032
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.7427
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

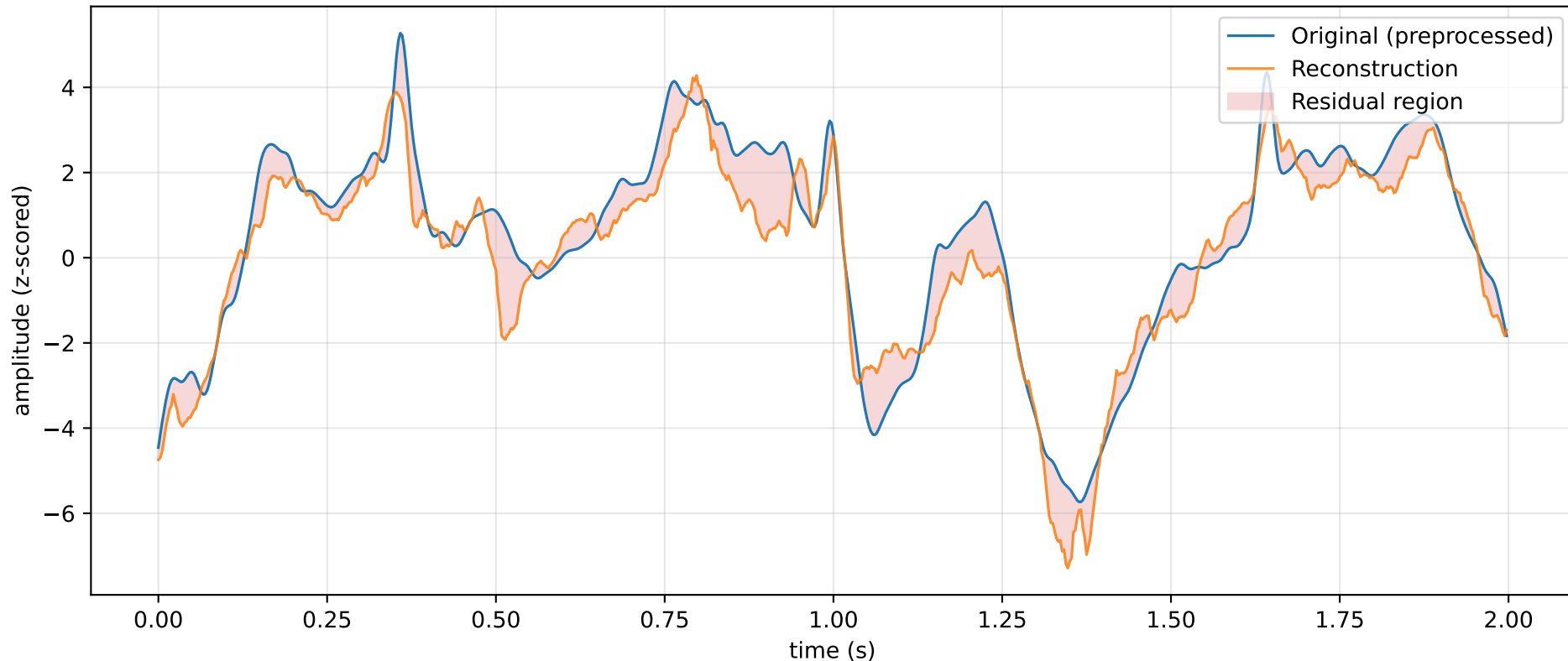
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 476122
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.7322
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

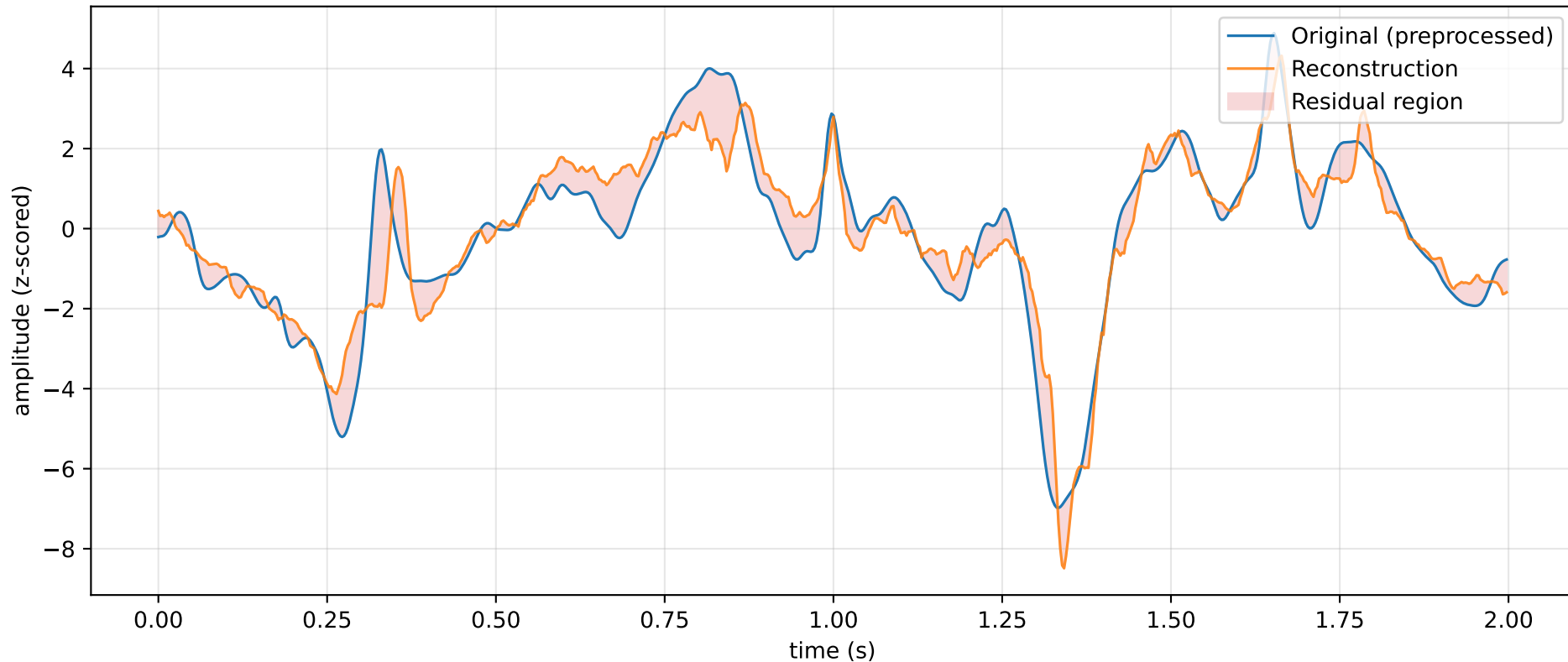
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 443710
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.7294
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

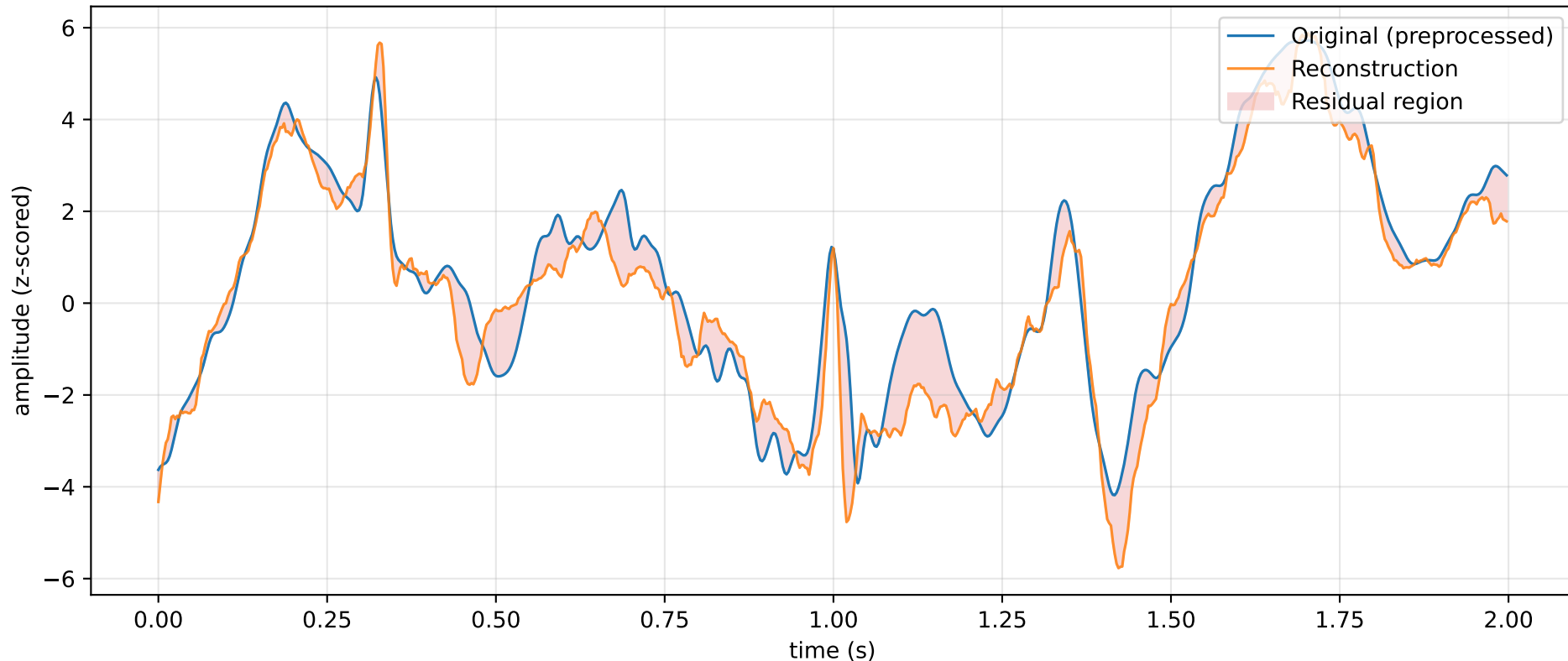
FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 475447
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.7276
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

FALSE POSITIVE — model wrongly flagged a normal beat: ground truth = Normal beat (N)



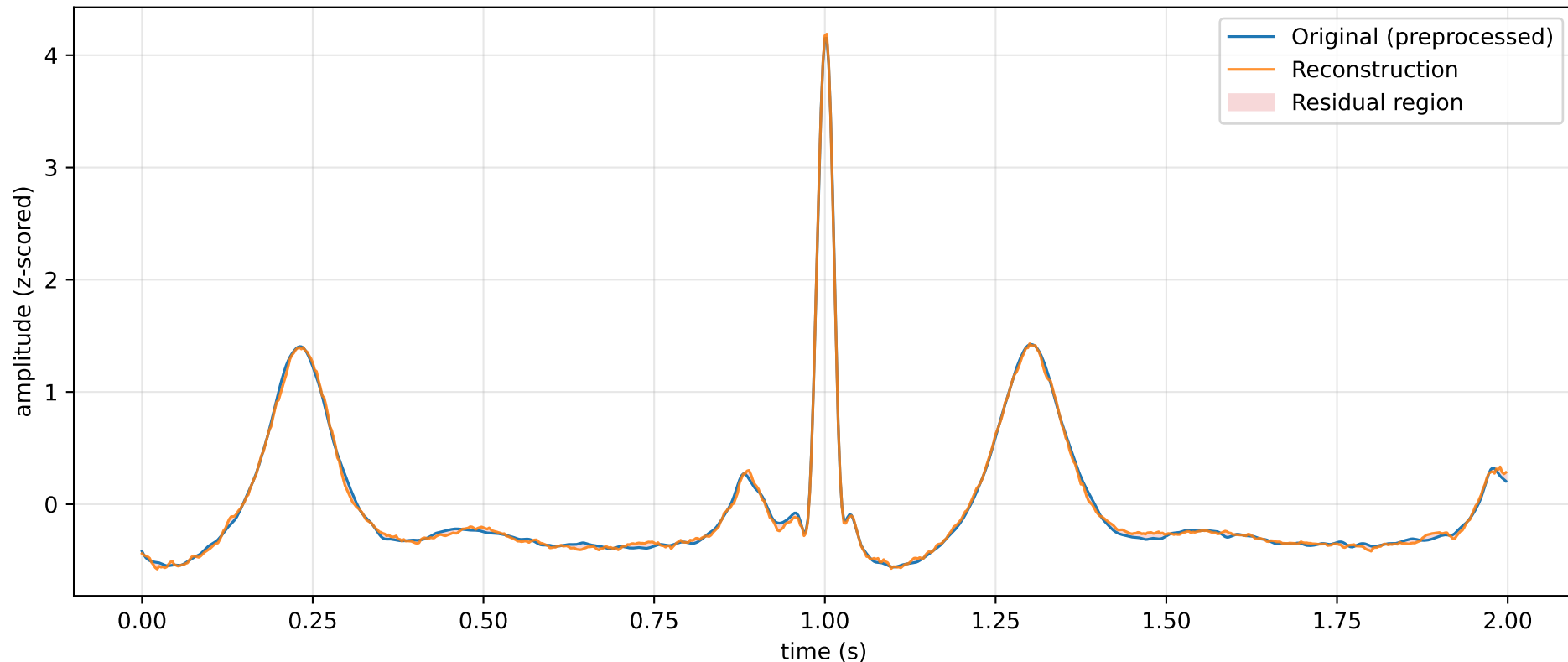
Source : rec 105, sample 433506
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.7122
Threshold : 0.0434
Model says : ANOMALY

ANOMALY

TRUE NEGATIVE — model correctly cleared a normal beat

(12 example(s) follow)

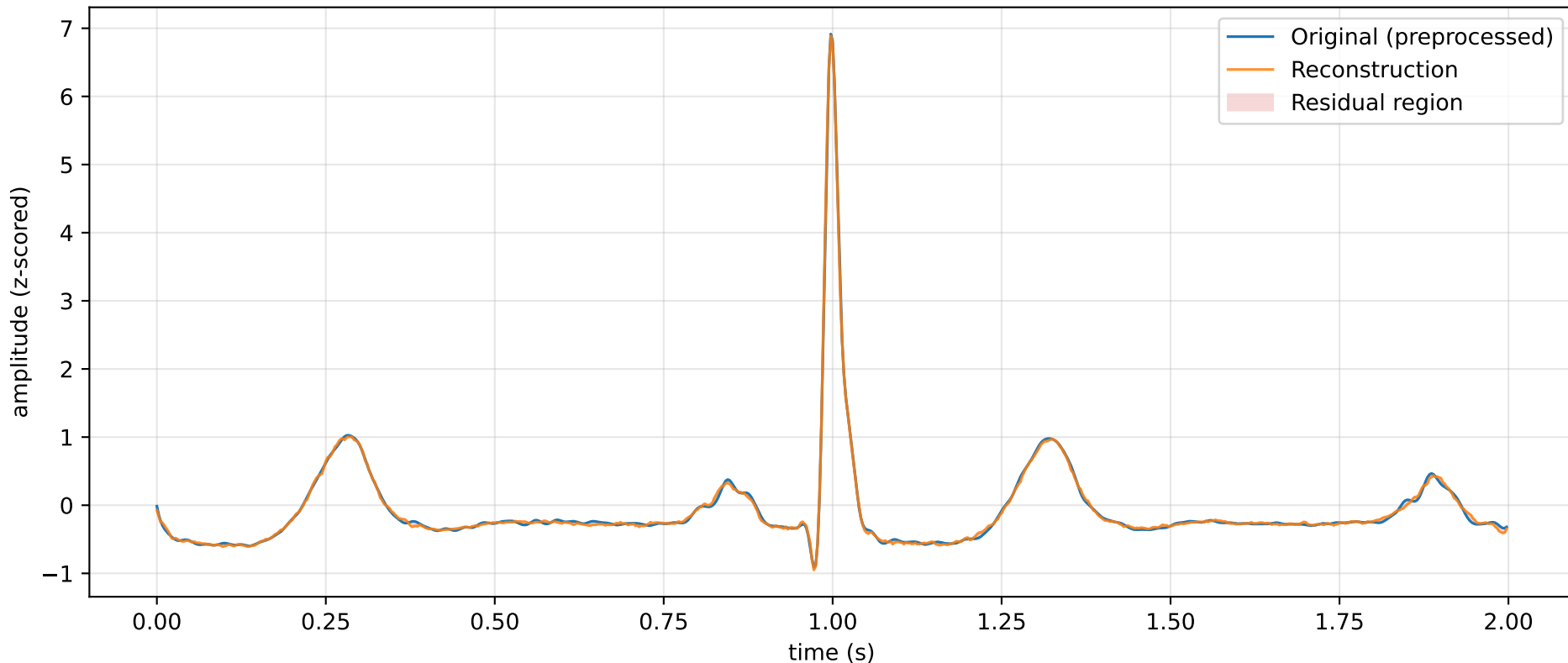
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 106, sample 109668
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0008
Threshold : 0.0434
Model says : NORMAL

NORMAL

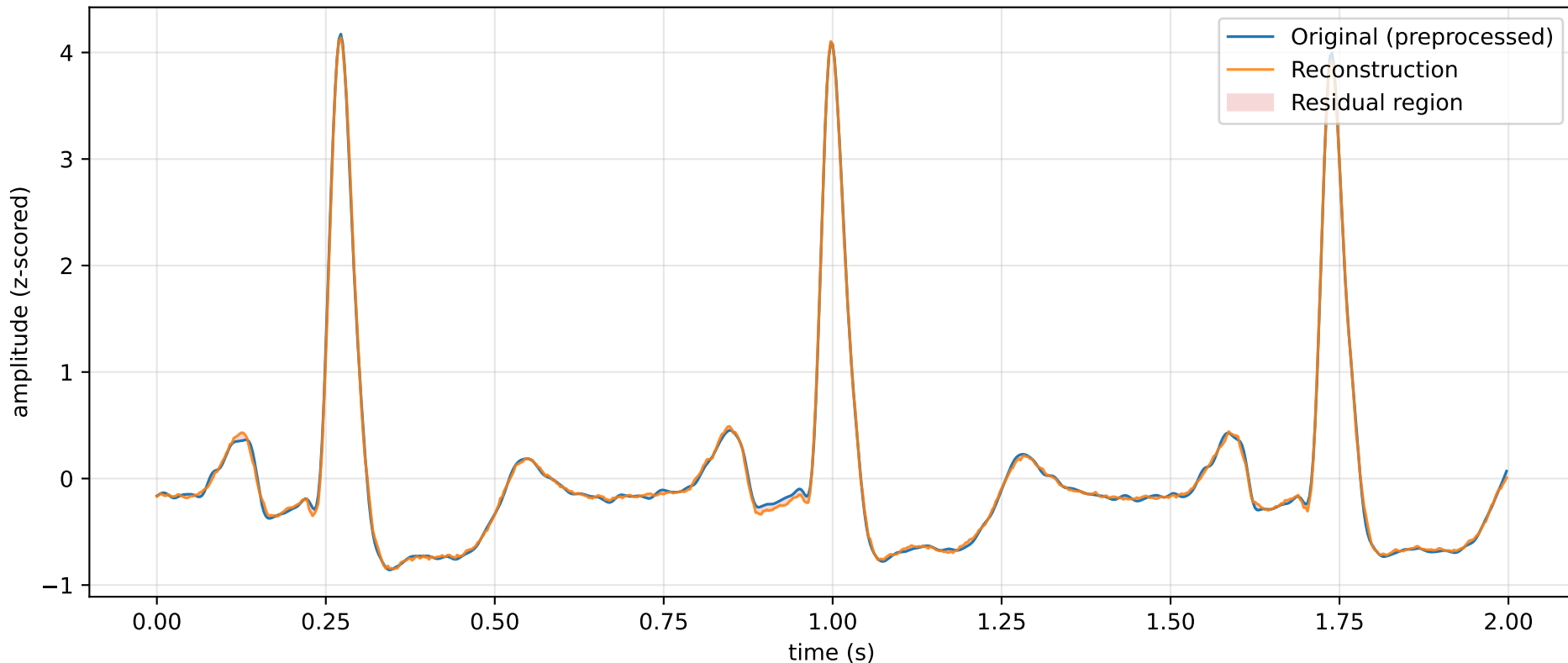
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 101, sample 385980
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0008
Threshold : 0.0434
Model says : NORMAL

NORMAL

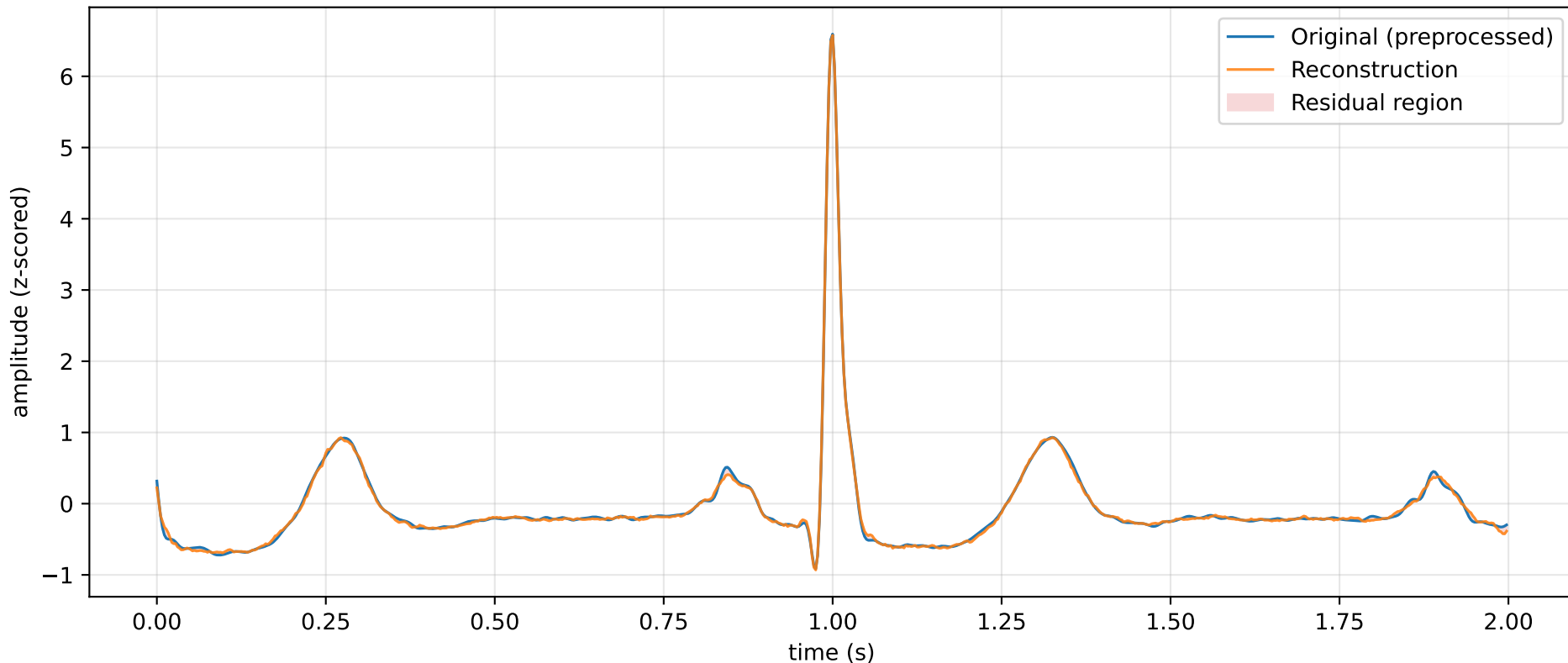
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 153530
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0008
Threshold : 0.0434
Model says : NORMAL

NORMAL

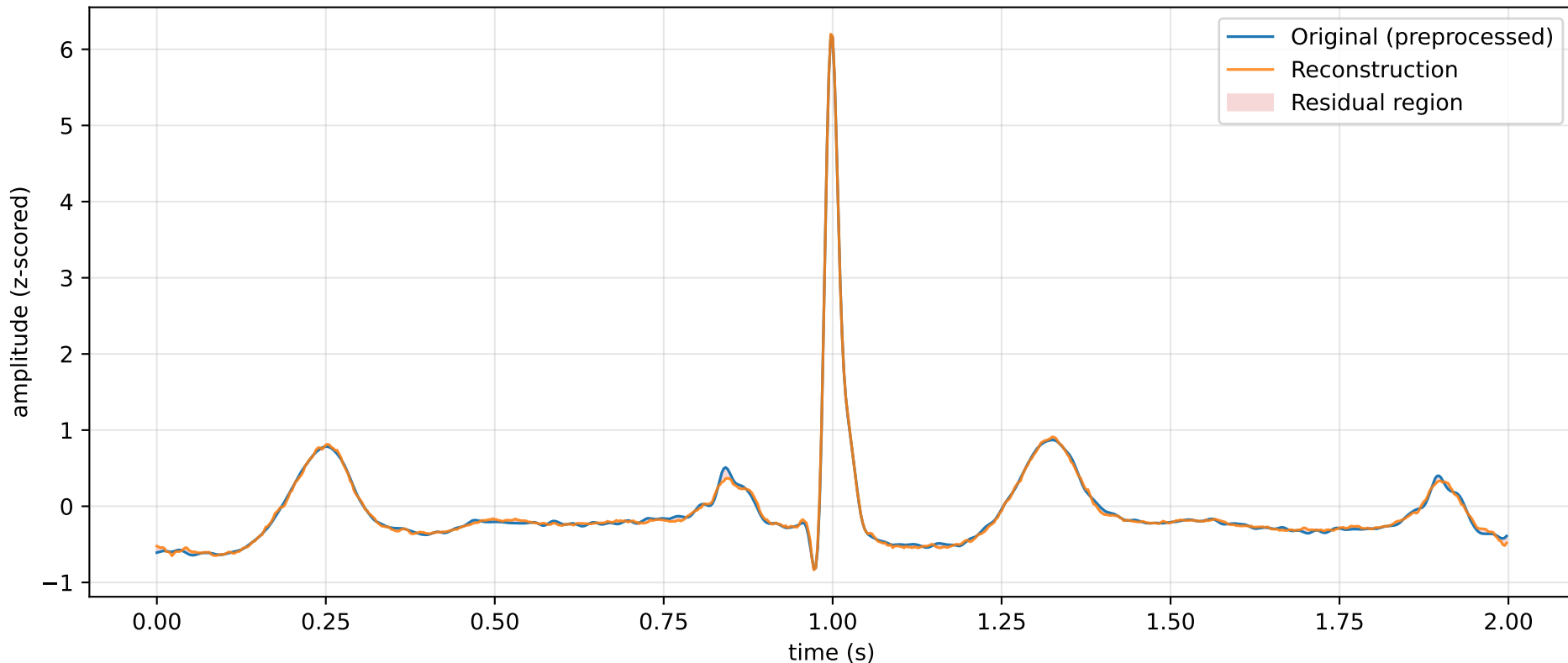
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 101, sample 331197
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

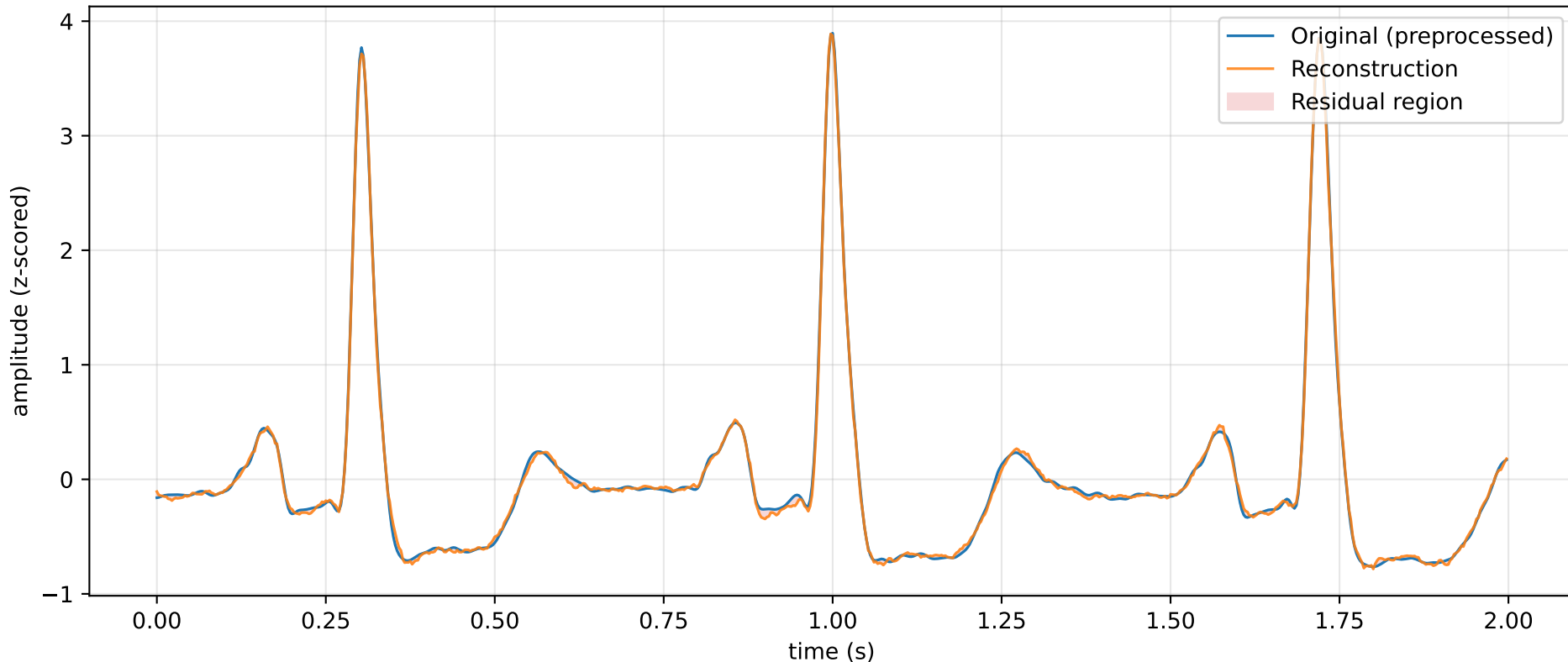
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 101, sample 303862
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

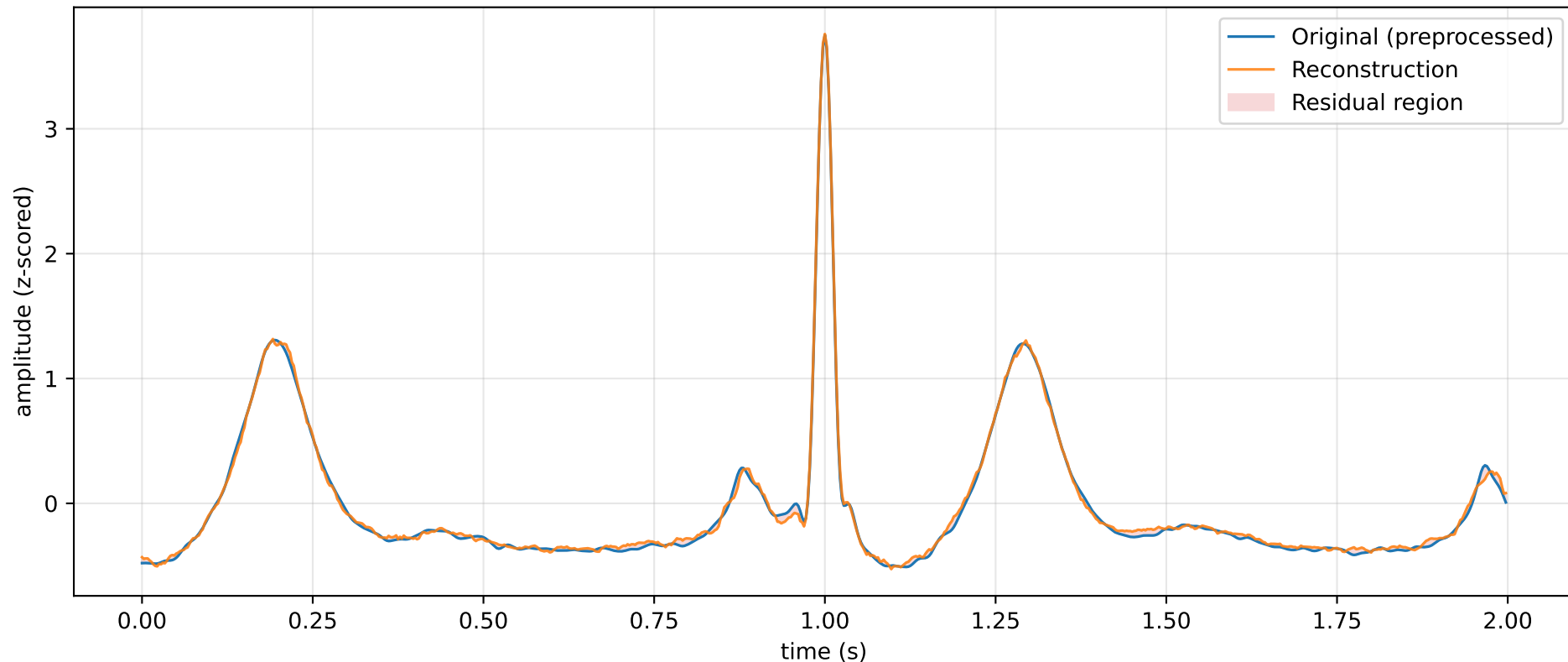
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 7246
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

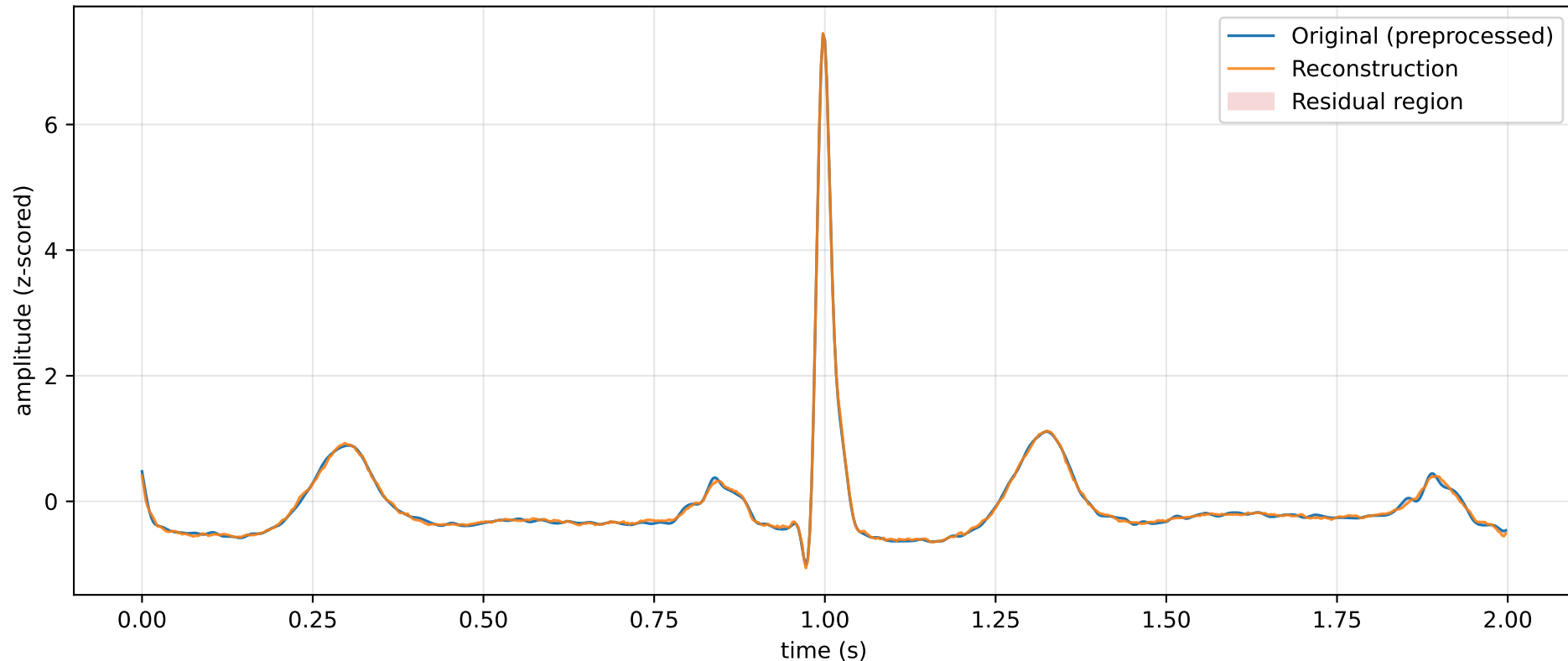
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 106, sample 105898
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

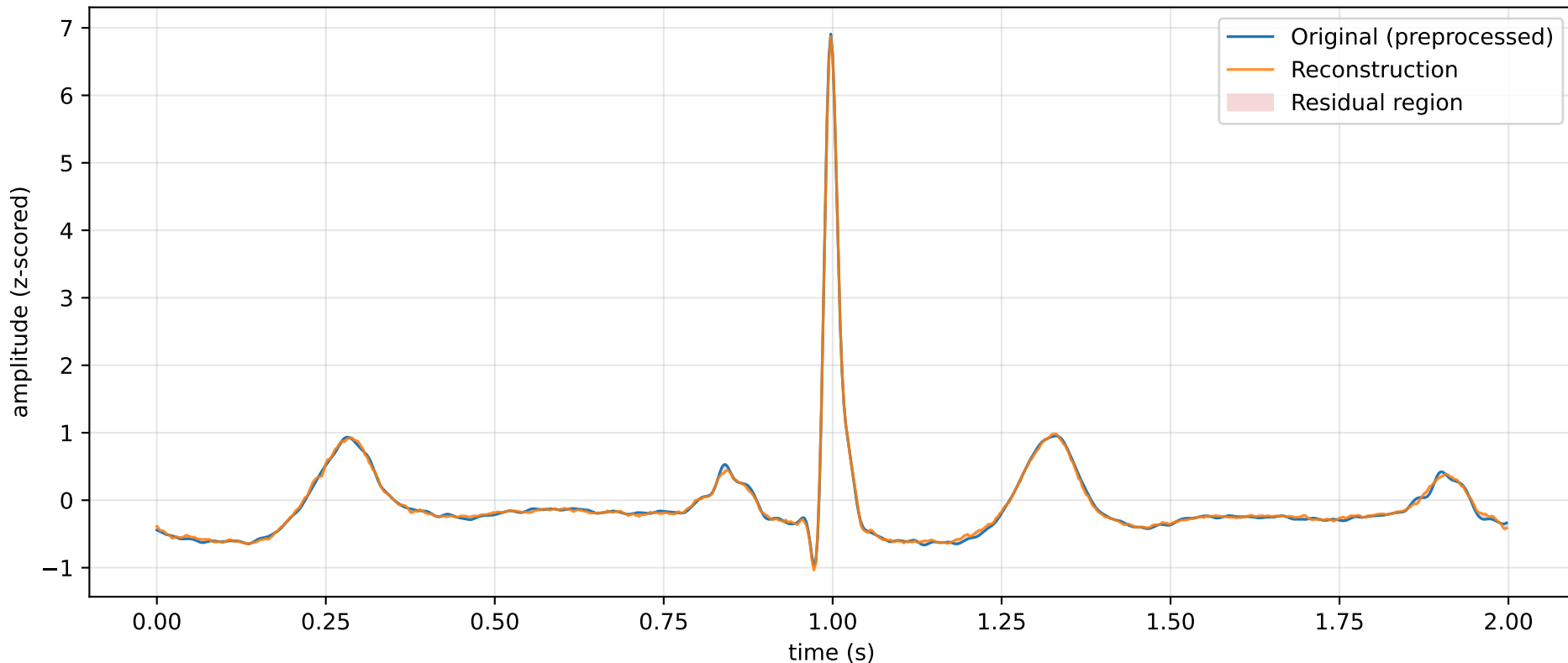
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 101, sample 422482
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

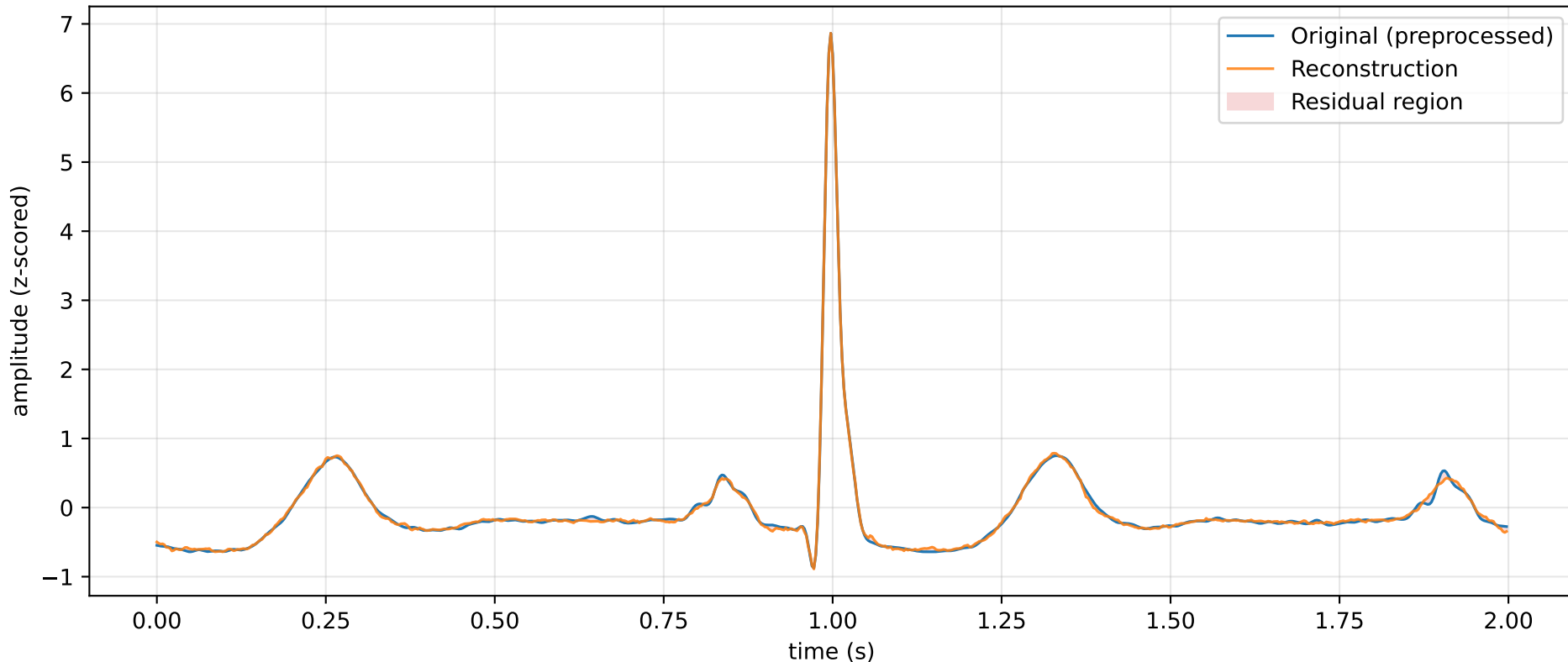
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 101, sample 254123
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

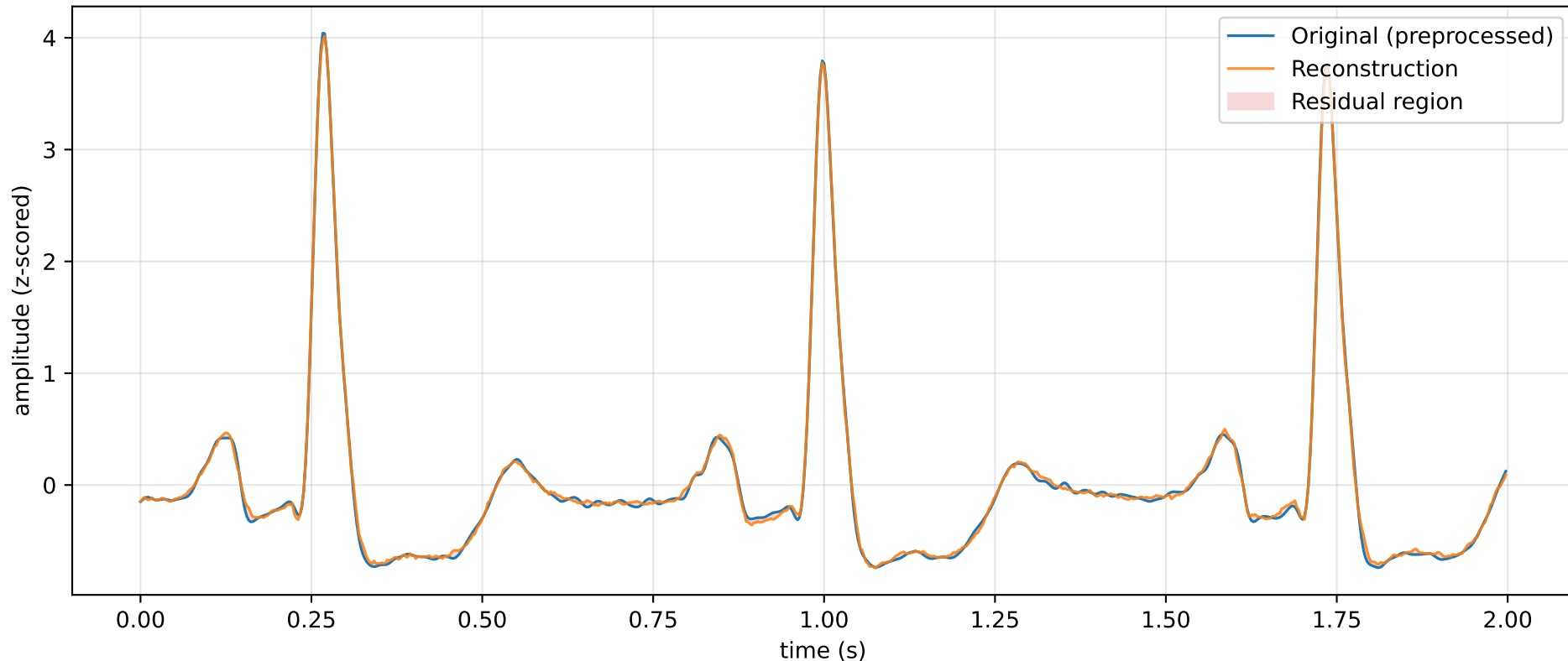
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 101, sample 630244
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

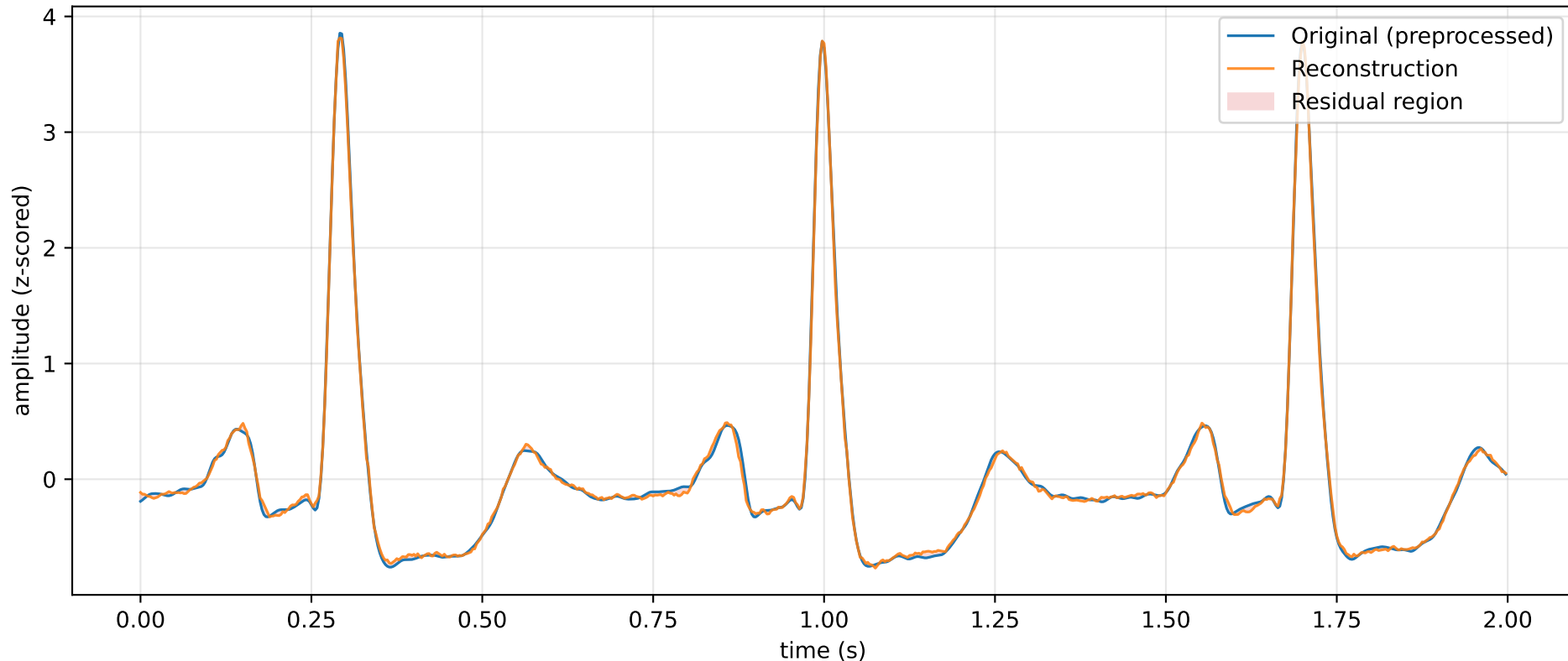
TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 131954
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL

TRUE NEGATIVE — model correctly cleared a normal beat: ground truth = Normal beat (N)



Source : rec 105, sample 20262
Beat symbol : N (Normal beat)
Ground truth : NORMAL
Residual : 0.0009
Threshold : 0.0434
Model says : NORMAL

NORMAL